Karnataka Climate Change Action Plan

An Interim Report submitted by

'Bangalore Climate Change Initiative – Karnataka (BCCI-K)'

to

Government of Karnataka

Submitted by

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Preamble

The Prime Minister's counsel on Climate Change announced in 2008 the need for India to formulate a National Action Plan on climate change. Eight National Missions were also constituted to form the core of the national action plan. The implementation of the plan would depend on 'institutional mechanisms suited for effective delivery of each individual Mission's objective and include public - private partnership and civil society actions'. As a part of this initiative, the Ministry of Environment asked all State Governments to prepare state level action plans.

A comprehensive climate action plan requires green house gas inventory, mitigation options, adaptation, stakeholder consultation and emergency and disaster management. These studies require credible data collection, strong analytical skills for data analysis using computational models and inter – disciplinary skills in science, and social sciences.

As of now, there are virtually no comprehensive and rigorous studies on the Karnataka situation examining the issues mentioned above, barring one on the Western Ghats. Realising the importance of the issue for the state of Karnataka, Prof B.K. Chandrashekar brought together distinguished experts and practitioners to establish the 'Bangalore Climate Change Initiative – Karnataka (BCCI – K)'. Earlier, as Chairman, Karnataka Legislative Council, he had set up in 2007-08 a State level steering committee on climate change. Several members of BCCI-K have worked on aspects of the Climate Change programme. These experts come from several prestigious institutions in Karnataka such as: Indian Institute of Science, Institute for Social and Economic Change, University of Agricultural Sciences, and Center for Study of Science, Technology and Policy. BCCI-K has also had support in the form of intellectual collaboration as also the benefit of comparative experience in climate change work across several countries.

The result of our efforts and studies will be made available to the Government of Karnataka and which can be used by the Government in the preparation of the state action plan. Our efforts have been adequately supported by the World Bank. Dr.Muthukumara.S.Mani and Dr.S.Vaideeswaran of the World Bank have also been of considerable help in shaping the analytical studies.

In particular, we wish to express our gratitude to the Hon'ble Chairman, Karnataka Legislative Council, Sri.D.H.Shankara Murthy; Chief secretary, Sri.S.V.Ranganath, IAS; Addl. Chief Secy. Smt.Meera Saxena and Secy., Forest and Ecology, Sri.Kanwar Pal for continued encouragement and for very valuable suggestions.

BCCI – K undertook the exercise of preparing the state action plan in two phases. We have now completed the Phase I, which involves GHG inventory, climate variability, and preliminary assessment of mitigating options and impact of climate change and vulnerability. This work was done under constraints of time and hence does not claim the analysis, rigorous as we believe it is, based on original empirical information from the various sectors. We hope to carry on these activities in Phase II (April – September). Phase II will consist of bottom-up studies to assess the impact of climate change at local level as well as vulnerability assessment at District, Forest-type and Watershed level. The second phase will also come up with specific mitigation and adaptation projects and prioritized through stakeholder consultation at the regional level.

We hope, again, that our studies will add value to the Climate Action Plan that will be formulated by the State Government.

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Chapter 1: Karnataka: Greenhouse Gas Inventory

Anthropogenic emissions of carbon dioxide (CO2) weighted by global warming potentials, constitute by far, the largest part of the emissions of greenhouse gases. Of these CO2 emissions, those that are produced from fuel combustion make up the great majority. CO2 emissions from burning biomass that a majority of rural households do for cooking is not considered, as biomass is considered to be carbon neutral.

In this section, we assess Karnataka's emissions of Greenhouse gasses (GHG) namely Carbon dioxide (CO2), methane (CH4) and nitrous oxide (N2O) at the state level. The methodology here used was similar to the 'GHG Emissions 2007', report from Ministry of Environment and Forests (MoEF)[1], which in turn has followed IPCC 1996[2] methodology.

The simplest way to estimate CO2 emission from fossil fuel combustion is assuming that the carbon in the fuel is released into the atmosphere in the short or the long run. Short-term emissions are defined within the *IPCC Guidelines* as those occurring within twenty years of the fuel use and are almost entirely reported in the fuel combustion module. Long-term CO2 emissions result from the final oxidation of long-life materials manufactured from fuel carbon and are usually emissions from waste destruction[3]. This methodology is called the top-down approach. In the more detailed bottom-up approach the computation of carbon released to the atmosphere is done at the plant level where combustion takes place taking into account the type of fuel used. The CO2 inventory in this report is calculated using top-down approach.

The sectors considered here are:

- Power Sector
- Transportation
- Residential/Commercial Buildings
- Industry
- Agriculture
- Waste
- Land Use and Land Use Change and Forestry (LULUCF)

[1] India: Greenhouse Gas Emissions 2007, MoEF, May 2010
[2] IPCC 1997
[3] http://www.ipcc-nggip.iges.or.jp/public/gp/bgp/2_1_CO2_Stationary_Combustion.pdf

Roughly, the annual total emissions from Karnataka will be around 80.2 million tons of CO2 equivalence (eq). LULUCF constitutes a negligible amount of net emissions national[4] and the same is assumed for the state. Here it should be highlighted that across sectors information was not available uniformly. While some data was for 2007-08 others were for 2009-10. Assuming there has not been any major shift in activity in these sectors, the number given is a fairly good estimate. The emissions from the three GHGs can be further broken down into the following:

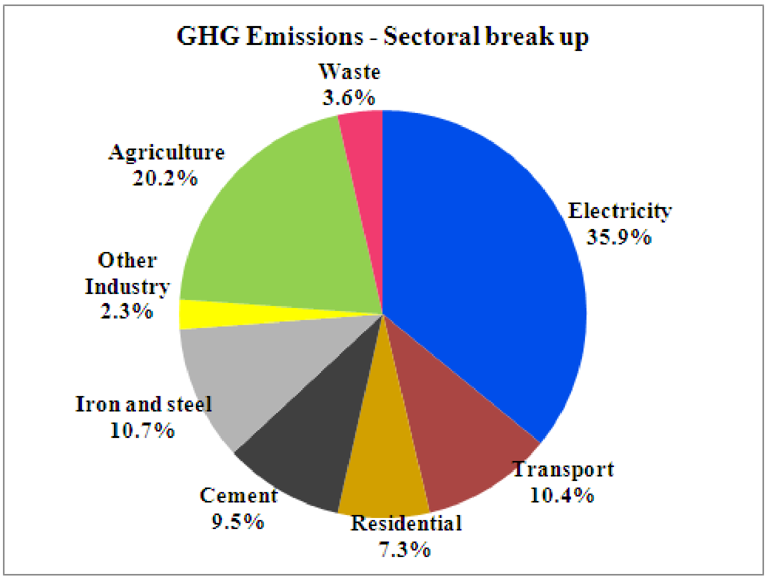
- CO2 emissions were 44 million tons (corresponding national number is 1221.7 million tons)
- CH4 emissions were 0.9 million tons (corresponding national number is 20.6 million tons)
- N2O emissions were 0.01 million tons (corresponding national number is 0.24 million tons)

In each of the cases, Karnataka's emissions are around 4% of the national emissions. In computing the GHG inventory, it was found that the granularity of information available was not uniform. For some sectors or sub-sectors, due to lack of state level information national statistics are used as a proxy. Some assumptions are made due to lack of data. For example, high speed diesel (HSD) is used for road transportation, diesel based electricity generation, irrigation pumps and for industries. However, while the total diesel sold in the state is known, the allocation between the sectors is not known, certain approximations are made and at times the national statistic are used as a proxy.

For example, it is known that there are very few diesel based irrigation pumps in the state and hence emissions from this source are assumed to be zero. Around 55% of the HSD sold in the country is used for transportation[5] and the same percentage is assumed for the state. It is assumed that the rest, 45% of the HSD sold is used by industry, although we recognize that this may not be the true picture. For example, the amount of HSD used diesel generators in commercial and residential buildings is ignored and the use by transportation might be different than the national statistic. The allocation of the emissions from each sector can be seen in the figure below and after which the details of the emissions and the methodology adopted are described.

[4] Per 2004 NATCOM report, this is true for India, and the same is assumed for Karnataka
[5] TEDDY 2010, TERI Energy Data Directory

Figure 1: Karnataka - GHG Emissions



Power sector

In the power sector, the net generation after factoring auxiliary consumption can be accurately calculated within a small margin of error. From the Central Electricity Authority's[6] CO2 inventory of coal, gas and diesel based plants, the CO2 intensity due to the generation from these sources was obtained. This was used to calculate the CO2 emissions in this sector. The total CO2 emissions from this sector were around 27.8 million tons for the year 2009-10[7]. Methane and nitrous oxide emissions are marginal here and are less than 0.5% of the total emissions in terms of CO2 equivalence; and therefore have not been accounted for.

Transport

Here the emissions from road, aviation and navigation are included. The total emissions due to this sector are calculated from the fuel consumed by vehicles used for road and air transportation. The total petroleum products sold by fuel type are known. While petrol sold in the state is exclusively used for transportation, aviation turbine fuel (ATF) is used by only airlines. However high speed diesel (HSD), is used for transportation, electricity generation by diesel generators and for running irrigation pumps. Light diesel oil (LDO) and furnace oil (FO) are used for transportation and by industry. A total of 8.35 million tons of CO2 are emitted by the sector as a whole. Here again CH4 and N2O are not accounted for.

[6] Central Electricity Authority, CO2 Baseline Database, Version 5, November 2009
[7] Installed capacity data from the Ministry of Power, Annual report 2010

Residential and Commercial Buildings

The emissions by the electricity consumption have been accounted for in the generation and hence are not considered here. However, the emissions due to cooking by liquid petroleum gas (LPG), kerosene and biomass are considered. Over 75% of the households in India use biomass. While biomass is considered carbon neutral their CH4 and N2O emissions are taken into account. As per national statistic it is assumed that 27.7 kgs of biomass is used per person per month. The total population using biomass is assumed to be 5.95 crores[8]. This implies roughly 20 million tons of biomass is used annually resulting in CH4 emissions of 0.09 million tons and N2O emissions of 0.012 million tons. The CO2 emission due to the use of LPG and kerosene is around 3.57 million tons. The total emission due to this sector is 5.84 million tons of CO2 equivalence.

Industry

Here the production volumes of the resource were obtained from various sources[9] and using emission factors given in the IPCC guidelines the total emissions were computed. The total emissions due to cement and iron & steel are high at around 8 million tons each. These two sectors are energy intensive and Karnataka's production volume of these resources is high.

Agriculture

GHG emissions here are mostly from methane and are largely due to livestock enteric fermentation and rice paddy cultivation. Livestock manure management results in the emission of N2O although this is a smaller percentage of the overall emissions by this sector. Given below in table 1 is a summary of emissions from this sector.

Table 1: GHG Emissions from Agricultural Sector

	CH_4 emissions (in million tons)	**N_2O emissions (in million tons)**	**CO_2 equivalence (in million tons)**
Enteric Fermentation	0.50		10.54
Livestock Manure management	0.04	0.0065	2.93
Rice cultivation	0.13		2.75
Total	**0.68**	**0.0065**	**16.23**

[8] *Population projection for India and states 2001-2026, Census of India 2001*
http://nrhm-mis.nic.in/UI/Public%20Periodic/Population_Projection_Report_2006.pdf

[9] Directorate of Economics, Government of Karnataka, India Stat, Ministry of Petroleum and Natural Gas, Federation of Indian Mineral Industries (FIMI)

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Emissions from Livestock

The total livestock population in Karnataka in the year 2003 was 25.6 million as per the 17th livestock census[10]. This includes cattle, buffaloes, sheep, goat and pigs. Emissions here are due to two factors – enteric fermentation and manure management, and tier II approach of IPCC methodology was used in computing the actual emissions here.

However, the Indian feeding standard of livestock is vastly different than the underlying assumption in calculating the IPCC default emission factors. Hence, emissions factor arrived at for India specific Indian feeding standard was used. This was developed by Central Leather Research Institute (CLRI)[11].

The drawback in this analysis is more recent data was not available and livestock population for more recent year was not projected. It should be noted here that this methodology might be different than what was adopted by MoEF. The MoEF report seems to have used NATCOM report[12].

Rice Cultivation

Approximately 1.4 million hectares of land was under rice cultivation in the year 2007. However, the breakup of the land under different water management practices (or irrigation) was not available. Hence the national statistics was used as a proxy to arrive at this for the state. Continuously flooded water management practice is the most inefficient in terms of methane emissions. Multiple aeration of paddy fields has the lowest emissions. Rice cultivation in Karnataka resulted in a total emission of 0.13 million tons of CH4 or 2.75 million tons of CO2 equivalence. Here emissions from agricultural soil and field burning of crop residue have not been done so far.

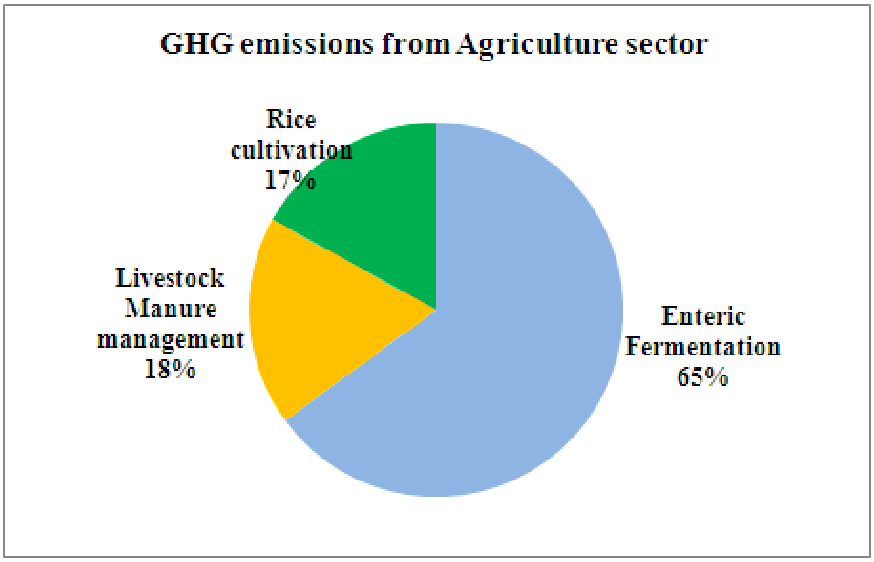
Table 2: Summary of CH_4 emissions from Rice cultivation

Ecosystem	Water regime	Rice Area 2007 (000' ha)	Methane Emission ('000 tons)
Irrigated	Continuously flooded	206	33
	Single Aeration	273	18
	Multiple Aeration	285	5
Rainfed	Drought Prone	115	8
	Flood Prone	309	59

[10] 2003 all India 17th livestock census
[11] *Budgeting Anthropogenic GHG emission from Indian livestock using country-specific emission coefficients,* Current Science, Mahadeswara Swamy and Sumana Bhattacharya, Vol 19. No.10, Nov 25, 2006
[12] NATCOM, 2004. India's Initial National Communication to the UNFCC. MoEF, GoI.

Deep water		42	8
Upland		168	0
Total		**1,396**	**131**

Figure 2: Break up of GHG emissions in Agricultural sector



Land Use Land Use Change and Forestry (LULUCF)

As per NATCOM 2004report, the emission from LULUCF is considered to be negligible. Hence it is assumed to be negligible for Karnataka as well and thus not included in the study.

Waste

The major green house gas emission from waste management is methane. The three main sources are from municipal solid waste, domestic waste water and the industrial waste water. The CH_4 emission is due to decomposition of waste in anaerobic condition and N_2O emissions from domestic waste water is due to its protein content.

In calculating CH_4 emissions from municipal solid waste management, only waste in urban area is accounted whereas in the rural areas the waste is scattered and does not come under anaerobic condition. IPCC (1996) revised methodology is adopted to calculate methane emission from municipal solid waste. The urban population in Karnataka is considered to be 37% of 2011 projected state population[13].The municipal waste generation rate on an average in all cities is estimated to about

[13] Mckinsey Report and population projection- Census India

0.55kg/capita/day and 70% reaches the land fill site[14]. These parameters are considered in the calculation of emission of methane, and found to be 69 thousand tonnes.

The waste water produced in Karnataka is estimated to 1036 million liters per day[15]. From the adopted IPCC 2006 guideline, it is calculated that 19 thousand tones of CH4 is emitted. However, the N_2O emissions are negligible. Since there is a dearth in data on industrial waste water, we made approximations to arrive at the methane emission.

However, the overall emission from this sector has been estimated as 5% of the national emission from the waste sector which accounts for 2.89 million tons of CO2 equivalence emission in Karnataka. More detailed analysis is needed to calculate the real emission from this sector.

Table 3: Summary of GHG Inventory*

	Quantity/ Production/ Area	**Production (units)**	**CO_2 emissions (in million tons)**	**CH_4 emissions (in million tons)**	**N_2O emissions (in million tons)**	**CO_2 equivalence (in million tons)**
Energy						
Electricity Generation (2009-2010)	11,495	MW	28.76	-	-	28.76
Transport (2007-08)			8.35	-	-	8.35
Residential (2007-08)			3.57	0.0900	0.0012	5.84
Industry (2008-09)						
Cement Production	120.97	Lakh tonnes	7.64	-	-	7.64
Iron & Steel Production	115.4	Lakh tonnes	8.59	-	-	8.59
Ammonia	2.36	Lakh tonnes	0.19	-	-	0.19

[14] National Environmental Engineering Research Institute, NEERI, 2005. Assessment of status of Municipal Solid Waste Management in metro cities, state capitals, class -I cities and class – II towns.

[15] Housing condition in India NSS report July-December 2002, http://moef.nic.in/modules/others/?f=event

Production						
Aluminium Production	1.09	Lakh tonnes	0.180	-	-	0.18
Iron Ore	423.14	Lakh tonnes	0.291	-	-	0.29
Pulp and Paper	3.65	Lakh tonnes	0.38	-	-	0.38
Sugar	33.97	Lakh tonnes	0.82	-	-	0.82
Agriculture						
Enteric Fermentation (2003)	2,56,17,000	Animals	-	0.50		10.54
Livestock Manure management (2003)	2, 56,17,000	Animals	-	0.04	0.00654	2.93
Rice cultivation(2007)	1.40	Million hectare	-	0.13		2.75
Waste			-	0.126	0.00079	2.89
Total						**80.16**

*It has to be pointed out again that the data for the same period was unavailable for all sectors. All attempts were made to get the most recent data for each of the sector. Hence it is likely that the total emissions will be higher than the 80 million tons given above. One could have fixed a particular year and projected the emissions of sectors where data was unavailable. However, this would introduce more errors as several assumptions had to be made in the projection. Going forward the government should take concerted efforts to collect annual data.

Chapter 2: Climate variability and climate change projections for Karnataka

Executive summary: In this chapter, we first analyzed observed climate data (over the past 50 years) to study current climate variability for the various districts of Karnataka. Our main findings were:

- On an average, the annual rainfall over the state has decreased by 6% over the period of 1951-2004. The decrease is higher than average in the coastal and northern parts of the state.
- Temperatures have increased by all over the state with higher increases (≥ 0.6°C in the last 100 years) in the northern regions.

Secondly, we analyzed the projections for climate change for the state. The main findings are:

- All of the state is projected to experience a warming of 1.8 to 2.2 0C by the 2030s (relative to the 1970s).
- The annual rainfall for the north-eastern and south-western parts of the state are projected to decrease. This projection is consistent with decreases that are already being observed over the last 30 years.
- The north-eastern districts of the state are projected to experience more frequent incidences of drought.

1. <u>Introduction</u>

The starting point for an investigation of adaptation to future climate change is to develop an understanding on the adaptation to current climate variability. This means that the focus for the investigation of adaptation to climate change should not be only on scenarios of the future but also on the analysis of present vulnerability in the face of current climate variability. Rainfall and Temperature are subjected to variability on all time scales: intra-seasonal, interannual, decadal, centennial, etc. Food production system, water availability, water resources etc. are sensitive to intra-seasonal, inter-annual, decadal variability in these two important climate variables. Global warming and Climate Change is projected to increase the number of extreme temperature and rainfall events, and hence climate variability is expected to show an upward trend. It is very important to understand the past trends and variability in rainfall, minimum and maximum temperature in Karnataka since the knowledge on the past could provide guidance for the future.

2. Current Climate Variability in Karnataka

Climate variability refers to variations in the mean state (of temperature, monthly rainfall, etc.) and other statistics (such as standard deviations, statistics of extremes, etc.) of the climate on all temporal and spatial scales beyond that of individual weather events. Variability may be due to natural internal processes within the climate system (internal variability), or to variations in natural (e.g. solar and volcanic) external forcing (external variability).

In this section, we focus on the current mean climate and climate variability in Karnataka at district level and investigate how changes in them will alter Karnataka's vulnerability to climate change. Precipitation and temperature are used as the key climate variables in this analysis.

Data and Methodology

The high resolution (0.5° x 0.5° lat. and long.) daily gridded rainfall dataset for a period of 35 years (1971–2005) provided by Indian Meteorological Department (IMD) for precipitation and the Climatic Research Unit Time Series (CRU TS) version 2.10 on a 0.5° lat x 0.5° l long resolution monthly dataset spanning 102 years (1901-2002) for temperature were used.

District-wise data was obtained by re-gridding the dataset to 0.1° lat x 0.1° long, and re-aggregating by the districts to study the climate variability at district level.

2.1 Rainfall Variability

- The IMD gridded rainfall analysis shows that there is a long-term negative trend in precipitation over Karnataka for the period 1951-2004 (Figure 1(a)).
- There is considerable decrease in precipitation over the Coastal and North Interior Karnataka districts (Figure 1(b)).

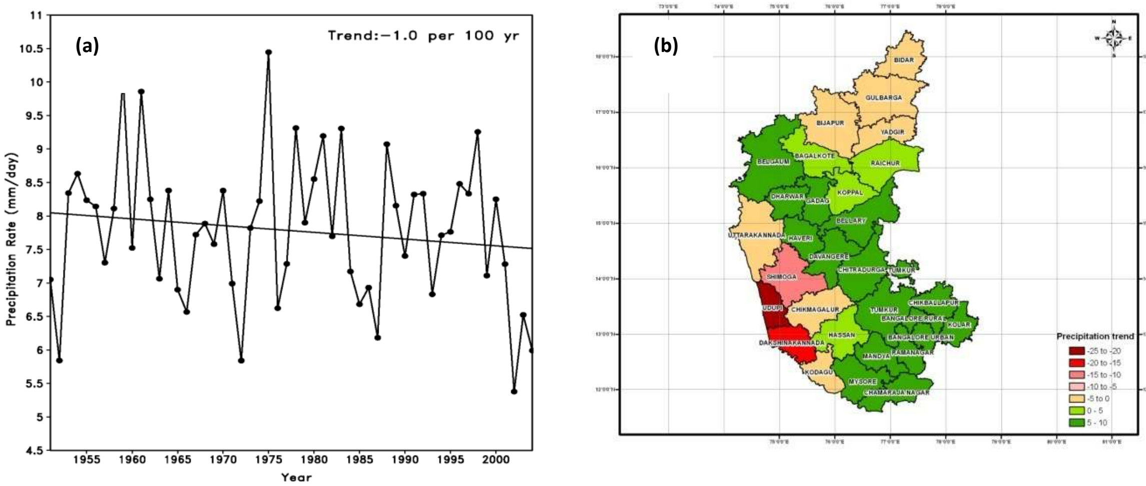


Figure 1(a): Time series of precipitation rate during monsoon season (June-September) over the box 11.5-18.5° N, 74–78.5° E in the IMD gridded rainfall analysis.

Figure 1(b): District-wise precipitation trend (mm/day per 100 yr) of southwest monsoon season (June-September) for the period 1971-2005.

2.2 **Temperature Variability**

The analysis of the meteorological measurements of temperature for Karnataka shows a steady warming trend in both the minimum and maximum temperatures (Fig.2).

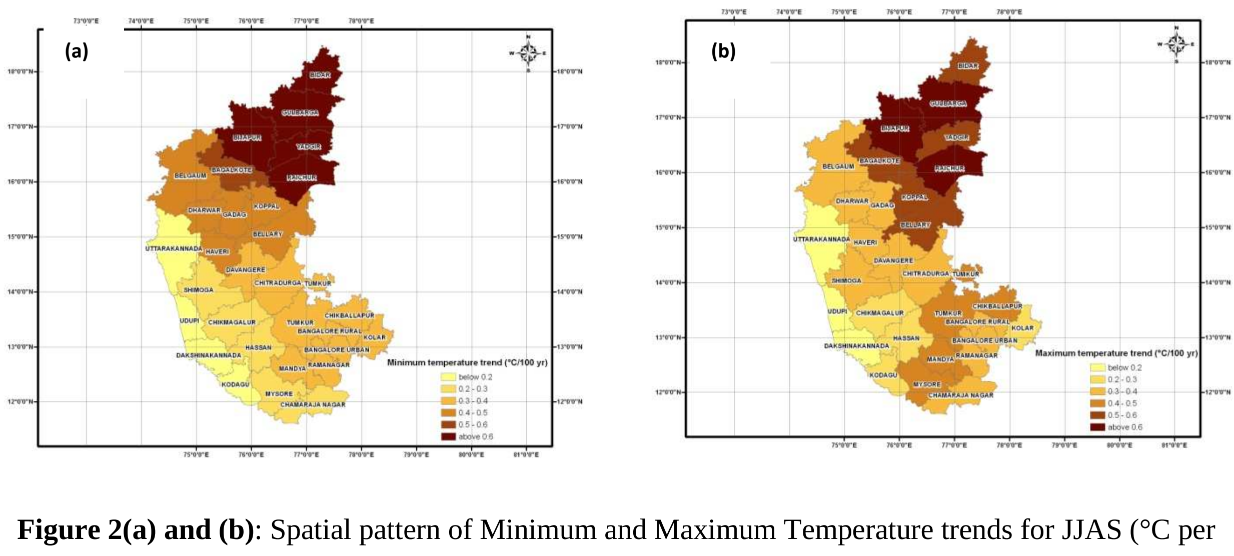


Figure 2(a) and (b): Spatial pattern of Minimum and Maximum Temperature trends for JJAS (°C per 100 yr) over Karnataka for the period 1901-2002.

- The spatial pattern of minimum temperature trend (Figure 2(a)) shows an increase all over the region. Districts of North Interior Karnataka (Bidar, Bijapur, Gulbarga, Yadgir and Raichur) experience an increase in minimum temperature of ≥ 0.6°C in the last 100 years. .
- The maximum temperature trend (Figure 2(b)) indicates an increase of ≥ 0.6°C in the last 100 years over Gulbarga, Bijapur, Gulbarga, Bidar, Raichur and Yadgir.

3. Future climate projections for Karnataka

3.1 Model and Methods

For climate change projections, we have used simulation data from the global climate model, HadCM3 from the Hadley Centre, UK (Collins et al., 2001). HadCM3 has been used recently for generating climate change projections for various parts of the Indian subcontinent (Kumar et al., 2006).

GCM and SRES scenario used: In this report, data from the HadCM3 global climate model downscaled by PRECIS model, a regional climate model for downscaling climate projections (see Kumar et al., 2006), is used. The combination of HadCM3 and PRECIS models is known as the HadRM3 model. The pathways for atmospheric greenhouse gases (e.g. CO_2, CH_4, N_2O, CFCs) were prescribed from the SRES A1B mid-term (2021-2050) projections. Climate change projections were made:

- For daily values of temperature (average, maximum, minimum)
- For daily values of precipitation
- At grid-spacing of 0.44250 latitude by 0.44250 longitude
- For periods of 2021-2050

Derivation of district-wise data: Data derived from the PRECIS model outputs (which had a grid spacing of 0.4425° latitude by 0.4425° longitude) was regridded to 0.2° in latitude and 0.2° in longitude. This ensures that enough grids fall inside each district. Then, the data was re-aggregated (as averages) at the district-level.

3.2 Projected increase in average, minimum and maximum temperature

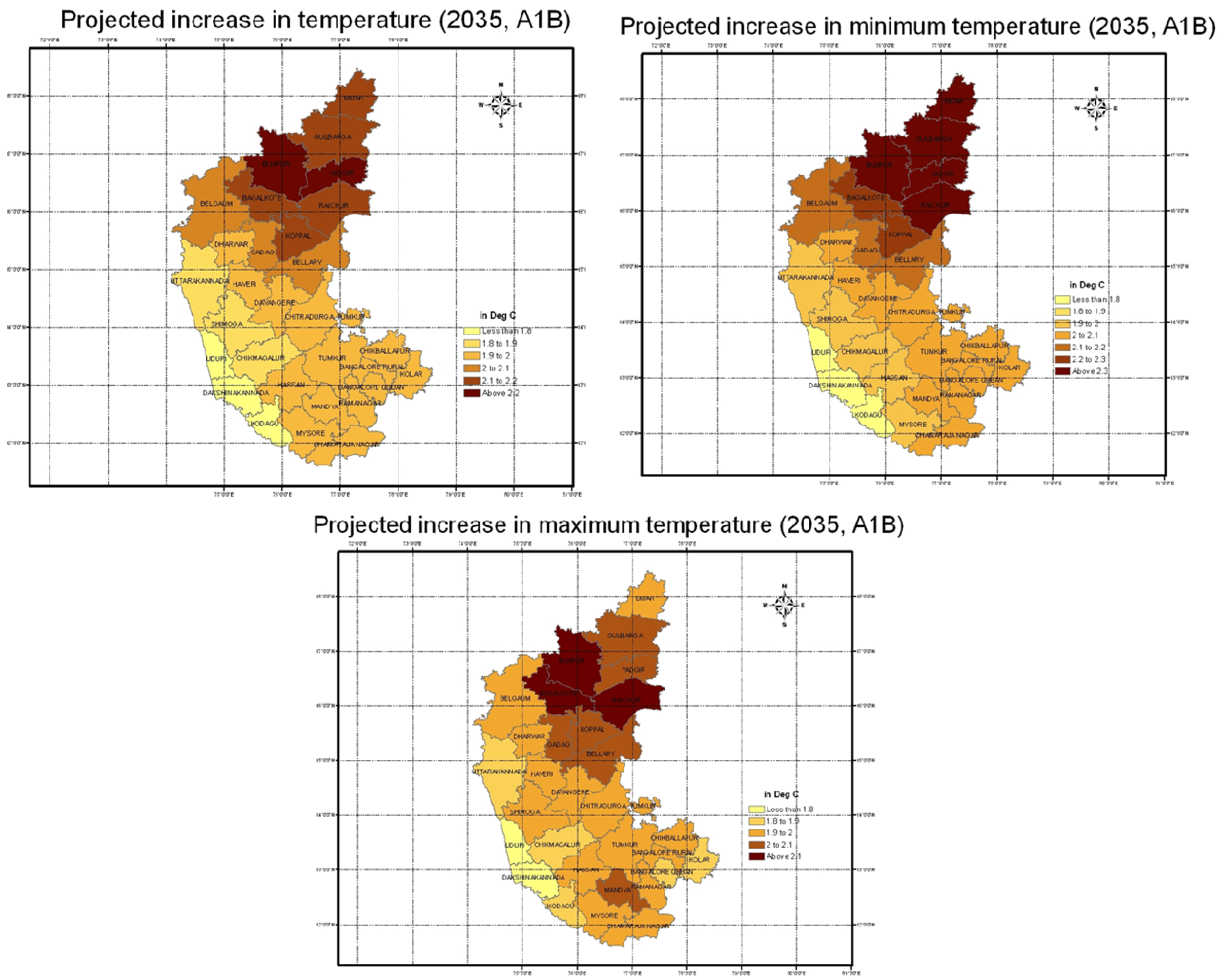


Figure 3: District-wise projected increase in annual average temperature (°C), minimum and maximum temperatures for the period 2021-2050 (A1B SRES scenario) compared to baseline (1975), projected by the HadRM3 model. The solid black lines show the district boundaries.

Figure 3 shows the projected increase in temperatures in the state.

We note that:

- The projected increase for annual average temperatures for the northern districts is higher than the southern districts. These regions are expected to experience a warming above 2^{0} C by 2030s.
- Most of the state is projected to experience a warming of 1.8 to 2.2 ^{0}C.
- The northern part of the state also is projected to have higher increases in minimum and maximum temperatures.

- The increase in the minimum temperature projected is slightly more than that of the average and the maximum temperatures.

3.3 Projected changes in rainfall

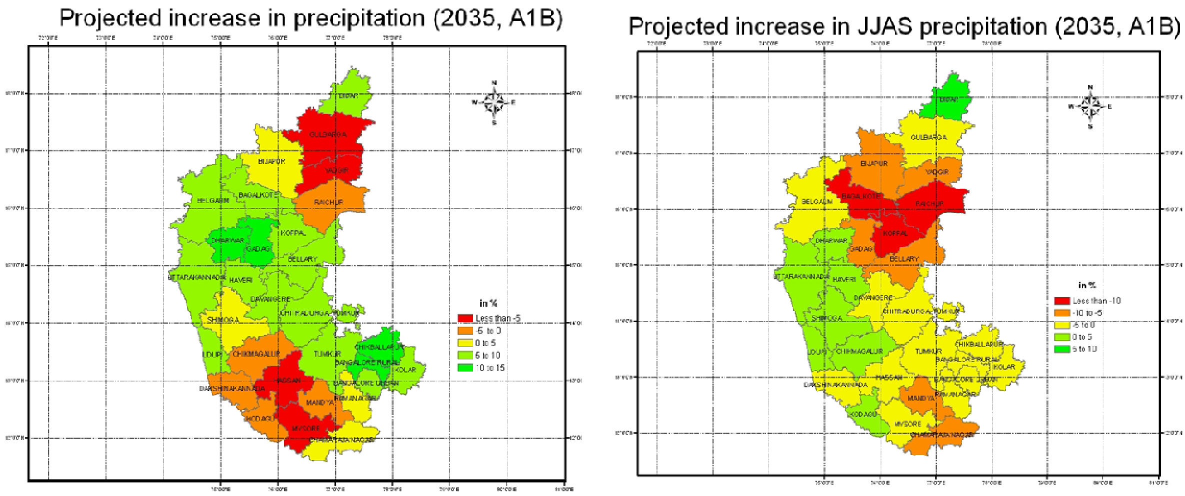


Figure 4: District-wise projected increase in annual rainfall (%) and JJAS rainfall for the period 2021-2050 (A1B SRES scenario) compared to baseline (1975), projected by the HadRM3 model. The solid black lines show the district boundaries.

Figure 4 shows the projected change in total annual rainfall and for the summer monsoon season (June, July, August and September months abbreviated as orJJAS).

It can be seen that:

- The north-eastern and south-western parts of the state are projected to experience decrease in the quantum of rainfall, annually. This roughly correlates with observed trends over the last 30 years.
- Over the JJAS season too, north-eastern and south-western parts of the state are projected to experience reduced amounts of rainfall

3.4. Monthly temperature increases for Bijapur

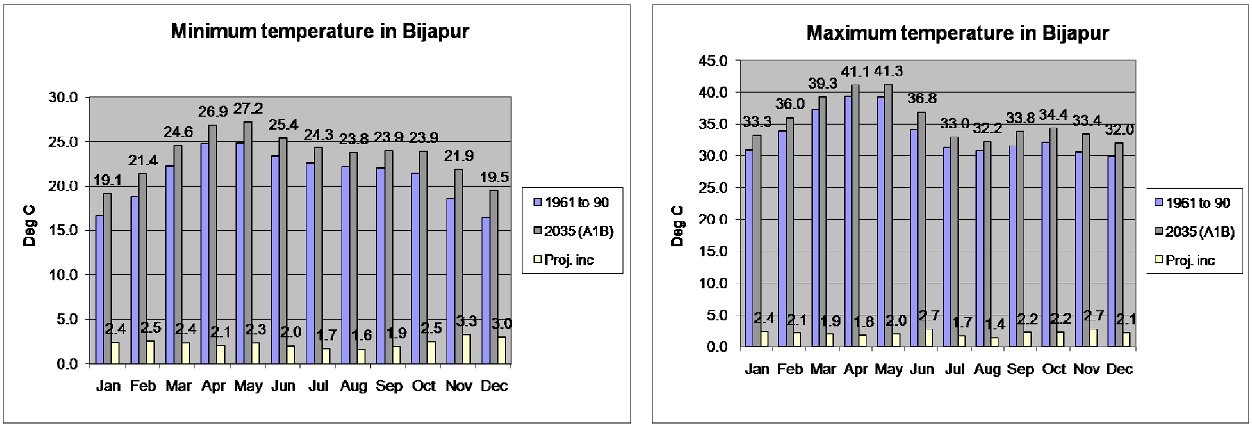


Figure 5: Projected increases in the maximum and minimum temperatures in Bjiapur.

We analyzed the projected increases on a monthly scale for all the districts. An example is given in Fig. 5 for Bijapur (a large and important district in the northern part of the state).

Salient observations are:

- The minimum temperature goes up by as much as 2.4 to 3.3 ^{0}C in the winter months (November to February)
- The maximum temperature goes up by as much as 2.7 ^{0}C in June and November.
- The average daily maximum temperature is projected to go up to as much as 41.3 ^{0}C in the month of May. This may be detrimental to various crop systems and natural ecosystems.

3.5 Projections for the frequency of droughts: We define a period of absence of rainfall (daily rainfall < 2.5 mm) for 40 or more contiguous days as "an incidence of severe drought". This agrees with the FAO definition of droughts. Further, we identify two major growing seasons in Karnataka: *Kharif* (July 2nd week to October 2nd week) and *Rabi* (September 1st week to February 4th week). The number of incidences of severe droughts are estimated for 2021-2050 for each grid point and compared to the baseline. The percentage increase is calculated; this increase gives an indication of possible increases in drought instances for that grid-point. Grids that are not drought-prone (number of severe drought instances less than 10 in the 30-year baseline) are excluded. The results of the analysis are shown in figure 6.

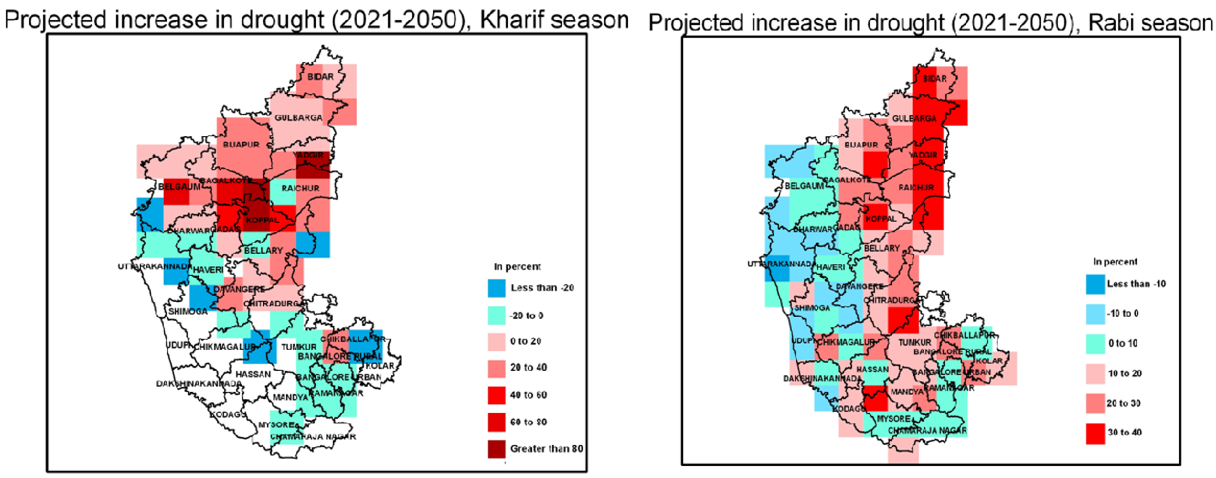


Figure 6: Projected increase in drought incidences in the future, compared to the baseline (1961-1990). The percentage increase in drought incidences is shown. The blue colors indicate that drought instances are projected to decrease, while the red colors indicate that such instances may increase.

We fint that:

- In the Kharif season, most northern districts are projected to have an increase in drought incidences by10-80%
- Districts of Koppal and Yadgir are projected to have almost a doubling of drought frequency in the Kharif season
- In the Rabi season, drought frequency is projected to increase in most of the eastern districts of the state.
- The western parts of the state are projected to have more rainfall and hence less number of droughts in the Rabi season.

4.0: Knowledge gaps and future work

- This study has used only projections from a single GCM namely HadCM3. Projections by the IPCC for global impacts of climate change typically use as much as 16 GCMs developed at centers from all over the globe. We should explore the possibility of using multiple GCMs so that climate change projections can be made more robust.
- We have used CO_2 concentrations for the SRES A1B scenario only. Exploration of other SRES scenarios (like A2 and B2) would provide us a range of possibilities for future.
- For this report, we have used climate projections generated at 0.4450 latitude by 0.4450 longitude. The accuracy of projections at district level is expected be higher at higher

resolution such as 0.10 latitude by 0.10 longitude. Future work should attempt to use final resolution data.

- For assessment of climate variability, we have used only two sources observational data: IMD and CRU. But meteorological data is being collected independently by various institutions around the world. Collecting and collating such data from them and building up such an integrated and robust database need to be explored.

References:

1. Collins, M., Tett, S.F.B., and Cooper, C. (2001). The internal climate variability of HadCM3, a version of the Hadley Centre coupled model without flux adjustments. *Climate Dynamics* 17: 61–81
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Chapter 3: Climate change and forest sector in Karnataka

Executive Summary: In this report, we first briefly outlined the state of forests in the state, and then described projections for the impact of climate change for forests in the state. Lastly, we detailed some policy recommendations in the context of the Greening India Mission. Our main findings are:

- There are around 3.62 Mha of forests in Karnataka. The area under forests is marginally declining in the state, while the area under dense forests has declined significantly (more than 8% from 2001 to 2007).
- It is projected that the forests of Uttar Kannada, Chikkamagalur and Shimoga districts are particularly vulnerable to the projected climate change even by as early as 2030s. Overall, around 38% of the forest area is projected to be impacted by climate change by the 2030s.
- The annual incremental mitigation potential of forestry in the state is estimated to be 15.5 million tonnes of Carbon or about 57 million tonnes of CO2 by 2020.

Introduction

Forest sector is important in the context of climate change due to three reasons namely; i) deforestation and land degradation contributes to about 20% of global CO_2 emissions, ii) forest sector provides a large opportunity to mitigate climate change, particularly the REDD (Reducing Emissions from Deforestation and Degradation) mechanism, and iii) forest ecosystems are projected to be adversely impacted by climate change, affecting biodiversity, biomass growth and forest regeneration. Climate is one of the most important determinants of vegetation patterns globally and has significant influence on the distribution, structure and ecology of forests. It is therefore logical to assume that changes in climate would alter the configuration of forest ecosystems. Based on a range of vegetation modeling studies, IPCC (2007) concluded that a-third of biodiversity is under the threat of extinction and further, significant impacts of climate change are projected on forest biodiversity and regeneration, especially in tropics and mountain areas. Studies at Indian Institute of Science (IISc) concluded that under the climate projections for the year 2085, 77% and 68% of the forested grids in India are likely to experience shift in forest types under A2 and B2 scenarios, respectively. A recent study from IISc using a dynamic global vegetation model showed that at the national level, about 45% of the forested grids are projected to undergo change by 2030s under A1B climate change scenario.

Thus, Government of India has formulated a National Greening India Mission in the context of climate change adaptation and mitigation and to sustain the ecosystem services of forests.

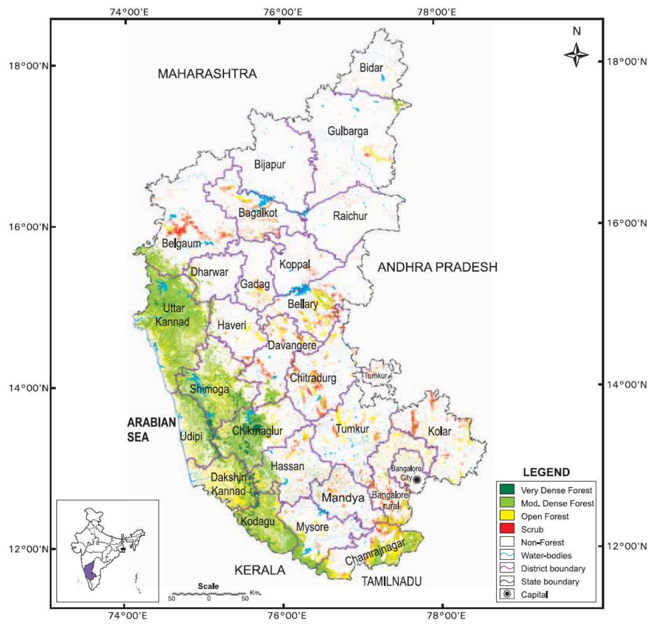
<u>Objectives</u>

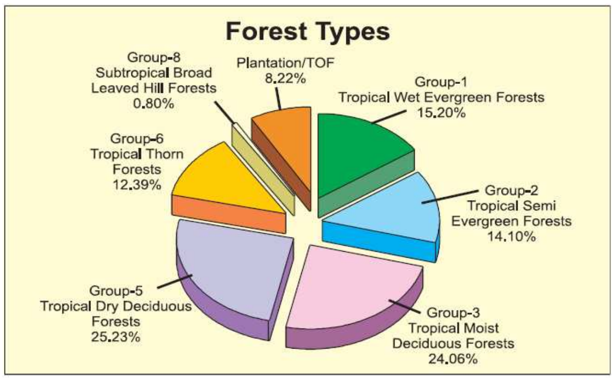
Forests account for about 19% of the geographic area of Karnataka. Karnataka is also home to bulk of the biodiversity rich Western Ghats. Thus the impact of climate change in Karnataka is of great significance and adaptation measures may be required to enable forests to cope with the climate change. Under the State Action Plan **firstly, the impact of climate change on forests of Karnataka is assessed using dynamic modeling and secondly, adaptation and mitigation options and projects are developed for forest sector of Karnataka in the context of Greening India Mission (GIM).**

<u>State of forests in Karnataka</u>

The distribution of forests along with crown densities in Karnataka is given in Figure 1a. The area under forests in Karnataka is estimated by Forest Survey of India at 3.62 million hectares, account for about 19% of the geographic area. Moderate dense forests account for about 56 % of the forest area followed by very dense forests accounting for only about 5%. Among the forest types (Figure 1b)) tropical dry deciduous and tropical moist deciduous forests account for about 25% each and tropical evergreen + semi evergreen forests together account for about 35%.

Figure 1: Karnataka Forest Cover Map (1a to the left) and Forest types of Karnataka (1b – to the right) (FSI, 2009)





Trends in area under forests are given in Table 1. According to Forest Survey of India, area under forests seems to have marginally declined during the period 2001 to 2007. The dense forest has declined significantly during this period and consequently the area under open forest has increased.

Table 1: Trends in area under different types of forest in Karnataka (sq km)

Forest type	**2001 assessment**	**2003 assessment**	**2005 assessment**	**2007 assessment**
Dense forest	26156	22461	21968	21958
Open forest	10835	13988	14232	14232
Scrub forest	3245	3141	3173	3176
Total	**40236**	**39590**	**39373**	**39366**

Afforestation and forest conservation programmes in Karnataka:

- Karnataka has implemented a large afforestation programme. Around 1.2 million hectares of degraded forest and non forest lands have been afforested in the past 25 years. Karnataka has adopted multiple models under the afforestation programme including; Ecological restoration through Natural regeneration, Assisted Natural Regeneration, Plantations for Timber production, Plantation for fuelwood and small timber production, NTFP Plantations, School forest, farm forestry and Restoration of Mangroves.
- More than 700 million seedlings have been distributed to the farmers, institutions and people for agro-forestry, farm forestry and homestead gardens in the last 25 years.
- Karnataka has adopted Joint Forest Program to involve local people in protection, planning and management of forests. Under the national afforestation program, FDAs have been constituted in the forests and the wildlife divisions.
- As part of Biodiversity protection and conservation initiative, around 17% of the forest area has been brought under protected area network (5 national parks and 21 Wildlife Sanctuaries).

Impact of climate change on forests of Karnataka:

Methods and Models: An assessment of the impact of projected climate change on forest ecosystems in Karnataka is made using the following:

- **Climate model**; Regional Climate Model of the Hadley Centre (HadRM3)
- **Climate change scenario**; A1B scenario
- **Climate impact model;** global dynamic vegetation model IBIS

- **Period of assessment**; short-term (2021-2050) and long-term (2071-2100) periods.
- **Input data**; monthly mean cloudiness (%), monthly mean precipitation rate (mm/day), monthly mean relative humidity (%), monthly minimum, maximum and mean temperature (C) and wind speed (m/s), soil parameter (percentage of sand, silt and clay) and topography.

Impacts of climate change: The dynamic global vegetation model has been validated by Indian Institute of Science for its suitability for Indian conditions. The impacts are assessed at regional climate grid scales (about 50kmx50km). It can be observed from Table 2 that during the short term period of 2030s, out of the 1946 forested grids **in Karnataka 747 (38%) will be impacted by climate change**. Similarly a higher percentage (62%) of the forested grids is projected to be impacted by climate change by 2080s. This means that the projected climate is not suitable for the existing forest types and the species present. The distributions of the forested grids which are projected to be impacted by climate change are presented in Figure 3 for 2030s and Figure 4 for 2080s. Forested grids mainly in the central and northern parts of Western Ghats and South-east are projected to be impacted by climate change.

Climate change and Western Ghats: Western Ghats is one of the biodiversity hotspots. According the dynamic global vegetation model projections, the forests of Uttar Kannada, Chikkamagalur and Shimoga districts are particularly vulnerable to the projected climate change even by as early as 2030s. Thus the **biodiversity rich Western Ghats region of Karnataka is projected to be adversely impacted by climate change threatening the biodiversity.**

Table 2: No of forested grids projected to be impacted by climate change during 2030s and 2080s in Karnataka

Period	**Number of grids affected**
Total number of grids	1946
Number of grids projected to be affected in 2035	747
Number of grids not affected in 2035	1199
Number of grids projected to be affected in 2085	1210
Number of grids not affected in 2085	736

Figure 3: Forest vegetation change projected by 2035 under A1B scenario in Karnataka

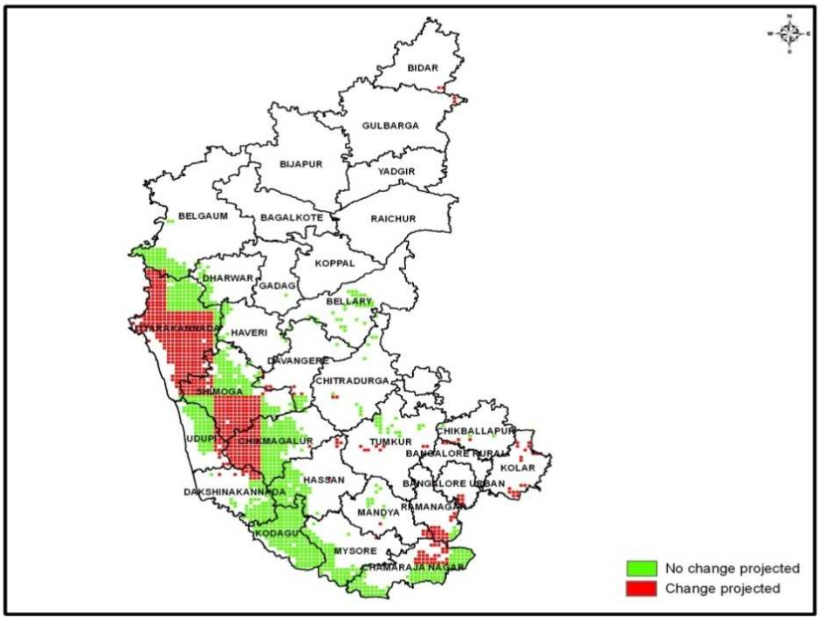
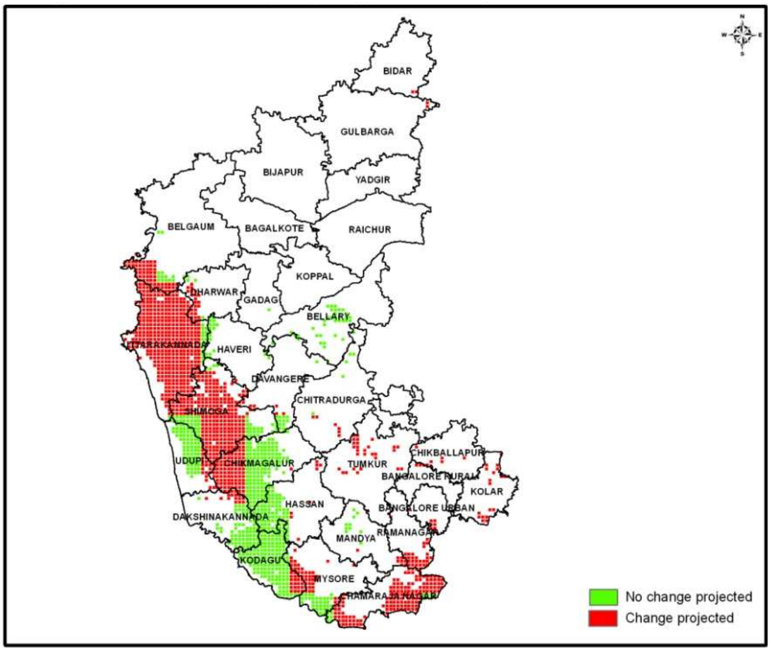


Figure 4: Forest vegetation change projected by 2085 under A1B scenario in Karnataka



Greening India Mission; Mitigation and adaptation

Government of India has formulated a large Greening India Mission aimed at mitigation and adaptation. Greening is meant to enhance ecosystem services such as carbon sequestration and storage, biodiversity conservation and provision of biomass and NTFPs. The mission aims at responding to climate change by combination of adaptation and mitigation measures which would aim at;

- Enhancing carbon sinks in sustainably managed forests and other ecosystems
- Adaptation of vulnerable species/ecosystems to the changing climate and
- Adaptation of forest dependent communities.

Thus under the state action plan both adaptation and mitigation projects are proposed for addressing climate change impacts on forest ecosystems as well as to mitigate the climate change through enhancing the carbon sinks**.**

Adaptation programme under the Greening India Mission: It was shown earlier in Figure 3 and 4 that significant proportion of the forests in Karnataka is vulnerable to climate change risks. There are no scientific studies to recommend specific adaptation measures suitable for different vulnerable forest types and regions. Studies by Indian Institute of Science have suggested some of the *win-win* adaptation strategies and practices to promote adaptation to projected climate risks. Table 3 presents a preliminary list of potential adaptation interventions and project ideas, particularly focused on the Western Ghats. There is a need for conducting preliminary studies to identify locations for implementing the adaptation measures. The cost of each of the adaptation interventions is not readily available. Many of the **adaptation interventions such as anticipatory planting, promotion of natural regeneration, mixed species forestry, and prevention of fire can become an integral part of the mitigation projects proposed under the greening India Mission**, listed in Table 4. Linking Protected Areas should be one of one priority projects under the adaptation programmes.

Table 3: Adaptation projects proposed for Karnataka under the Greening India Mission (GIM)

Category of Adaptation interventions	**Proposed adaptation Activities / Projects in Karnataka**
Anticipatory planting of species across latitudinal and longitudinal gradient	This will be a component of all mitigation programs / projects proposed under the Mitigation component of GIM (Table 4)
Promotion of natural regeneration and mixed species planting	This will be a component of all mitigation programs / projects proposed under the Mitigation component of GIM (Table 4)
Adoption of short rotation species	For all afforestation projects in the western ghats region
Effective fire prevention and fire management	For implementation in all the current fire prone areas of the state

Linking Protected Areas (PAs) management (for securing corridors for species migration)	Identify locations for linking existing PAs As a pilot project linking two PAs is proposed linking – Kuduremukh National Park and Krishnagiri Sanctuary
Reduced forest fragmentation by conserving contiguous forest patches	Identify locations for linking forest fragments in the western ghat region and a pilot project is proposed

<u>Mitigation projects under the Greening India Mission</u>: Forest sector provides a large opportunity for mitigation of climate change, in particular through reducing CO_2 emissions by reducing deforestation and forest degradation as well as increasing carbon sinks in the existing forests and creating new sinks in degraded lands through afforestation. The GIM has identified five sub-missions and several activities or interventions. The proposed mitigation programs and projects under the GIM are presented in Table 4, along with area proposed and the investment cost required.

Table 4: Mitigation projects proposed under the Greening India Mission (GIM)

GIM activities	***Categories under sub-missions***	***Regions of state***	***Total Area***		***Investment cost per hectare (in Rs.)***	***Total investment cost (in crores of Rs.)***
			Total Potential Area (in Mha)	***Extent of Proposed Area (in Mha)***		
1	2	3	4	5		
Sub-mission 1: Enhancing quality of forest cover and improving eco-system services	Moderately dense forest cover, but showing degradation	Western Ghats	2	0.2	15,000	300
	Eco-restoration of degraded open forests	Western Ghats and other areas	1.4	0.4	30,000	1200
	Restoration of grasslands	Outside forests	0.94	0.1	35,000	350
	Total		**4.34**	**0.7**		**1850**
Sub-mission 2: Eco-system	Restoring scrublands		0.32	0.1	50,000	500
	Restoration of mangroves		0.001	0.001	70,000	7

restoration and increase in forest cover	Restoration of abandoned mining areas		-	0.005	100,000	50
	Total		**0.32**	**0.106**		**557**
Sub-mission 3: Enhancing tree cover in Urban areas	Avenue, city forests, municipal parks, gardens, households, institutional lands, etc.,		-	0.02	10,000	20
Sub-mission 4: Agro-forestry and Social Forestry	Shelterbelt plantations		0.015	0.01	8000	8
	Private lands		5	0.5	20,000	100
	Highways/ rural roads/canals/tank bunds (ha)		0.045	0.03	70,000	210
	Total		**5.06**	**0.56**		**2745**

Mitigation potential of the proposed projects under greening India mission: Mitigation potential of proposed activities (Table 4) is estimated using COMAP model and Carbon sequestration rates used in the Greening India Mission. The annual incremental mitigation potential (Table 5) is estimated to be 15.5 million tonnes of Carbon or about 57 million tonnes of CO_2 by 2020**.**

Table 5: Incremental annual mitigation potential of proposed activities different options

Options	Area (Mha)	Incremental annual mitigation potential 2020 (MtC)	Incremental cumulative mitigation potential 2010-2020 (MtC)	Incremental cumulative mitigation potential 2010-2030 (MtC)
Moderately dense forests	0.2	1.8	13.7	32.0
Degraded/open forests	0.4	7.4	55.4	129.2
Scrub/grassland ecosystems	0.205	2.2	16.3	38.1
Mangroves	0.001	0.0	0.1	0.2
Agroforestry & social	0.56	4.1	30.7	71.7

forestry incl. urban forestry				
	1.366	**15.5**	**116.2**	**271.2**

Institutional arrangements for mitigation and adaptation programs

The **institutional arrangement** proposed for mitigation and adaptation programmes to address climate change proposed under the State climate change action plan is given in the box below.

Activities		**Institution**
Research	Carbon mitigation projects	Research wing of Karnataka Forest Department (KFD) and Indian Institute of Science (IISc)
	Impact and vulnerability modeling	Research wing of KFD and IISc
	Adaptation projects	Research wing of KFD, IISc, Institute of Social and Economic Change (ISEC) and Forestry College at Sirsi and Ponampet
	Long term monitoring	Research wing of KFD and Forestry Colleges at Sirsi and Ponampet
Monitoring	Carbon stocks	Research wing of KFD, IISc and Forestry Colleges at Sirsi and Ponampet
	Biodiversity	Research wing of KFD and Forestry College at Sirsi and Ponampet
	Growth rates	Research wing of KFD and Forestry Colleges at Sirsi and Ponampet
	Socio-economic aspects	Research wing of KFD and ISEC
Implementation	Overall the implementation of the mitigation and adaptation programmes under the Green India Mission would constitute an additional programme implementation responsibility along with the regulatory and developmental responsibilities that the KFD discharges at present. Forest Department does not have enough staff strength especially at lower levels to shoulder enhanced targets. Neither is the present set of staff adequately trained for a qualitative and people oriented Joint working. However keeping in view the above mentioned limitation, it is proposed to set up a special purpose vehicle totally different in composition and culture for effective implementation of Greening India Mission. At the department level, the works in the	

	notified forest areas will be taken up through the territorial wing of the department while those outside notified forests would be implemented through a separate *Directorate of Social Forestry* that will be assisted at Circle/Division level by subject matters specialist like Sociologist, Economist, Extension and Training Experts, etc for enhanced effectiveness. While the overall programme implementation will be facilitated, supervised and monitored by the KFD, the Village Forest Committees (VFCs) and Eco-development Committees (EDCs) will have a greater role in implementation of works at field level with the involvement of Non-Government Organization (NGOs) and other village level thematic groups like Self Help Groups (SHGs) under linkage with Gram Panchayaths. Research, modeling, GIS and Monitoring personnel need to be outsourced or engaged on contract basis or on deputation to have continuity of term so as to enable running of these facilities on professional lines. Such professional support is very essential to assist the Research, Working Plan and Evaluation Wings in the department.

Knowledge gaps and research needs: The climate change impact assessment is based only on one GCM (HadCM3) and only one Dynamic Global Vegetation Model (IBIS). Further, the plant functional types inbuilt in the model may have to be validated for the forest types present in Karnataka. Currently, only the impacts of climate change on forests are modeled. There is a need to develop vulnerability profiles of different forest types and forest divisions to enable identification and prioritization of the forest divisions and forest types for adaptation projects. Thus the following research priorities are identified:

1. To use multiple climate change model projections in impact modeling (preferably regionally downscaled) depending on availability.
2. In addition to IBIS, other DGVMs should also be used to assess the impacts.
3. Detailed assessment of plant functional types and existing forest type distribution will be made to assess the impacts according to existing forest types.
4. Vulnerability index will be developed and forest types and forest divisions or districts will be ranked based on the index to prioritize the regions for adaptation interventions.

Chapter 4: Agriculture sector in Karnataka State-Climate Change Impacts and Adaptation Strategies

Agriculture is a way of life for our people. In material terms, it provides sustenance for the vast majority of our population and accounts for nearly half of our national produce. On the success which attends our efforts here, depends also the success of our entire economic growth. Even now, much of our industry depends on agriculture for its raw material, exports emanate directly or otherwise from the agricultural sector and the growth of rural incomes will constitute an expanding market for the domestic industry.

Karnataka state forms the South Western part of the Deccan Peninsula and lies between 11.5°and 18.6° North latitude and 74.0° and 78.4° East longitudes. It is the 8th largest state in the country having an area of 191,791 Sq. Kms (6.25% of India's total area of 3,065,027 Sq.Kms.). The State has 30 districts, 176 taluks, 745 hoblies, 29,483 Villages. As per the census of 2001, the State has a total population of 5.27 crores accounting for 5.13 per cent of the country's total population of 102.70 crores. Sixty six per cent of the total population resides in rural areas, whose main occupation is Agriculture and allied activities. Out of the total population, 44.6 per cent is working population, of which 69.36 lakh are cultivators and 62.09 lakh are agricultural labourers. As per the Agricultural Census of 2000-01, the State has about 123.07 lakh hectares of cultivable area out of total geographical area of 190.50 lakh hectares, accounting for 64.60 per cent.

Rainfall situation of the State (1901-2008) and projected for 2035: The daily weather data has been obtained from IMD Pune and DES, GOK has been analysed. The State receives normal rainfall of 1152 mm of which monsoon receives 850 mm. Analysis indicated that the gradual declining in the south-west monsoon rainfall and decreased portion of rainfall is partly shared by Pre-monsoon and N-E monsoon. Hence there is a change in the rainfall pattern and amount of rainfall received during different months. In addition to this some part of the State likely to receive the s-w monsoon rains required for agriculture beyond July month (Ref. 1). The mean temperature is also having the increasing tendency.

Projected climatic variables for the year 2035 reveals the following observations: State is expected to receive 4.5 % more rains and likely to lose 3.1 % during south west monsoon season than what it is. Raichur, Koppal, Bellary, Bangalore, Bijapur, Chamarajanagar, Gadag and Mandya districts are likely to lose their s-w monsoon rains up to 12.5 %. Hence, in these districts only low water required crops like millets can be grown in rainfed agriculture. In other irrigated area regular irrigated crops can be grown. Bidar, Bijapur, Belgaum, Davanagere, Gadag and Gulbarga are the comfortable growing districts during s-w monsoon. DK, Udupi, UK districts from early June to early October, Davanagere, Haveri and Chamarajanagar districts from early July to late October,   Gadag and Gulbarga districts from early August to early October are the comfortable crop growing months. Bellary receives good rains only during August, Bidar to receive from early August to mid September, Chikmagalure and Shimoga districts from mid July to early October, Hassan district from early July to late October, Gulbarga and Raichur districts from Early August to early October, Koppal district from early August to early October, Kolar, Bangalore, Ramanagar Mandya and Mysore districts from late July to late October, and Tumkur and Chitradurga districts from mid July to late October.

The raise in mean annual temperature indicated that all most all the districts have indicated the raise up to 2.2 Degree Celsius except DK, Udupi and Kodagu districts that have indicated up to 1.8. UK, Mysore, Shimoga, Hassan and Chikmagalure districts have indicated to be warmer up to 1.9 Degree Celsius. This overall increase in temperature leads to increase potential evapotranspiration. Higher increase in temperature in the decreased rainfall districts causes quick loss of soil moisture and reduces the growing period. This also increases the drought duration causing low productivity. Winter and pre-monsoon months are indicating higher raise in mean temperature than s-w monsoon months. This leads to increased local convectional rains and thunder storm causing natural disasters. All most all the districts of the state barring Coastal, hilly and few eastern districts are likely to have increased wind speed. The increased wind speed may add to increase potential evapotranspiration causing further declining the growing duration.

<u>Procedure of Estimating the Impact of Climate Change</u>

Estimating baseline production*:* Simulations were done with InfoCrop models for major crops of the State using the respective crop coefficients and management for each of the 10 years from 1991 to 2000. The actual daily weather data of the respective districts were taken and simulated to get the predicted yield. The mean of 10 years realized yield was taken as the baseline yield. The actual crop yield in each district was compared with the predicted yield and the ratio (constant) of realized and estimated was generated for all the crops and for all the districts. These values were calibrated to district productivity and this was treated as the ‘baseline yields’.

Simulating production for 2035 scenario: Using InfoCrop models crop yield (Based on IPCC and national Climate Change Policies) was studied for major crops in the state. The climate models outputs on temperature (minimum and maximum) and rainfall for A1b-2035 scenario were used. Apart from this, frequency of occurrence of climatic extreme events such as higher/low rainfall events and high temperature events is inherently accounted for. The projected carbon dioxide levels as per

Bern CC model for scenario was also included in the model for simulations. Based on the simulated yields in changed scenarios, productivity was estimated as in case of baseline production taking the changed climatic scenario and enhanced CO2. Then the simulation outputs are multiplied with the ratio constants (generated using the baseline yields) to get the estimated yields. To express the impacts on productivity, the net change in productivity in climate change scenario was calculated and expressed as the percentage change from baseline mean productivity. The infocrop model was used for Maize, Sorghum, Rice, Redgram, Cotton, Potato, Soyabean and Wheat crops. The Stocheometric crop weather model generated at UAS, Bangalore (Ref. 2) was used to estimate the productivity in the case of Ragi. Similar procedure was adopted to generate the baseline yields and also to estimate the yield for the changed scenario of 2035.

Limitations of the Models: Both the models have been statistically tested with the observed and predicted productivities. The model is able to predict the yield up to 89 % (R^2) accurately.

Projected impacts of climate change on crops in 2035 scenario: The simulation analysis indicates that the productivity of kharif crops such as irrigated rice in the State is likely to change by -14.4 to 9.5% from its base yield. Majority of the irrigated rice growing area is projected to lose the yields up to about 8.2% and smaller non-rice growing districts projected to gain up to 6.2% indicating the net decline in the irrigated rice. In case of rainfed rice, the projected change in yield is in the range of -13.8 to 7.2% with large portion of the region likely to lose the rice yields up to 9.6%. Since irrigated rice, in general, is supplied with better amount of fertilizers than the rainfed rice, it has better opportunity to benefit from CO_2 fertilization effects for production and accumulation of dry matter and hence grain yield. Both Irrigated and Aerobic rice in smaller area of south-western Karnataka is likely to gain. In these areas, current seasonal minimum and maximum temperatures are relatively lower (20-22^{o}C Tmin; 27-28^{o}C Tmax). The projected increase in maximum temperature is also relatively less in these areas (1-1.5^{o}C) as also increase in minimum temperature, which is projected to be about 1.3^{o}C. In most of the Western ghats region, the monsoon rainfall is likely to increase marginally. These changes in temperature and rainfall cause direct impacts on the production of kharif crops as indicated in the study (References 2-4). Variable response in both irrigated and rainfed rice can be attributed to the influence of major limiting factor at a given location, viz., rainfall, temperature and nutrient supply. Increased rainfall can be a surrogate for reduced Sunshine duration due to cloudiness, causing limitation to photosynthetic rates resulting in to reduced accumulation of dry matter, particularly in areas with high rainfall. Since Western ghat receive high rainfall, it can be inferred that heavy cloud cover causing low radiation as one of the limiting factors for lower productivity in this region, any further increase in rainfall (thus more cloudiness) will result in reduced yields in kharif season. Farmers in Western ghat regions can reduce the impacts of climate change and can reap higher harvests by adapting crop management strategies such as soil moisture conservation, provision for proper drainage, increased efficiency of water and nutrient supply and utilization and by growing cultivars tolerant to adverse climate.

Climate change is likely to change yields of maize from 27.6 % to -19.3 % and sorghum by 17.2 % to -18.4 in different districts with respect to baseline yield. These crops have C4 photosynthetic system and hence do not have relative advantage at higher CO_2 concentrations. Increase in rainfall in already high rainfall zones is detrimental to the crop production as mentioned earlier. Further, increased temperature causes reduction in the crop duration due to increased growth rates. Reduced crop duration means less opportunity for crop canopy to accumulate the photosynthates and thus dry matter. These conditions can cause the reduction in gain yield as indicated in this analysis. Further, any coincidence of high rainfall with pollination period will affect the production to spikelet sterility, especially in cross pollinated crops such as maize and sorghum.

Marginal increase in rainfall may not hamper the Sunshine period in interior districts providing ample scope for crops to carry the photosynthetic process benefitting the rice yields in elevated CO_2 conditions. Currently the productivity of irrigated maize yields higher than that of rainfed maize. Maize, being a C4 crop, is able to accumulate higher biomass even at current CO_2 levels due to higher photosynthetic efficiency and thus yields high when water and other nutrients are not limiting. On the other hand, growth and yield of rainfed crops are limited by water and nutrients. In 2035 scenario, higher temperatures and reduced Sunshine because of more rains may limit the biomass production and yield as compared to baseline conditions. Even though C4 crops do not get direct benefit of elevated CO_2 for higher photosynthetic rates, they may benefit through leaf water balance. The estimated 2.1 °C rise in mean temperature and a 4.5% increase in mean precipitation would reduce net agricultural productivity in the state by 1.2. Agriculture in the coastal and ghat regions is found to be the most negatively affected. Small losses are also indicated for the major food-grain producing regions of few districts. On the other hand, interior North and South Karnataka districts have shown benefit in rainfed crops to a small extent from warming. The net change in the productivity of few major crops from the baseline yields in the state is indicated in the table 1. The vulnerable districts for few crops are listed in table 2. Although total productivity is likely to be decreased by about 1.2% in the State, due to the benefit of the raise in temperature and CO2 level, few districts have indicated gain in productivity up to 35% (in some crops). By changing over to such crops in those vulnerable districts, the loss in the productivity can be compensated and the advantages of the climate change effects can be absorbed positively.

Table 1. Changes in the Yield/Unit area under the changed climatic scenario of 2035.

District	Irri. Rice	Aero Rice	Maize	Sorghum	Red Gram	Cotton	Potato	Soya	Wheat	Ragi
	%	%	%	%	%	%	%	%	%	%
BAGALKOTE	4.6	5.6	1.4	5.4	-9	-2.3			-2	
BANGALORE - URBAN	3	1.7	27.6	10.1	14.1	22.6	-20.1	-0.8		-0.6
BANGALORE - RURAL	5.1	5.7	23.3	11.1	17.9	16.7		-11.4		-3.4
BELGAUM	-4.4	-4.1	13.2	4.9	-2	10.4			-5.8	2.3
BELLARY	-7.9	-5.9	-2	15.4	-8.3	-10				3.5
BIDAR	0.3	-6.9	8.3	15.3	1.5	9.1			9.8	2.3
BIJAPUR	3.4	-6.5	-9.1	2.9	-4.2	15.4			5.7	4.6
CHAMARAJANAGAR	-14.4	-5.8	12	10	9.1	2.9				-2.4
CHIKMAGALUR	-2.5	-7.9	-17.1	-6.7	17.6	-14.8	-16.2			6.8
CHITRADURGA	8.9	5.3	8	15	34	-6.3		-10		8.7
DAKSHINA KANNADA	-9.2	-9.6	-15.8	-10	-20	-16				-9.4
DAVANAGERE	5.5	2.8	7.1	-8.7	12.5	7.7	-8.8		3.4	-6.5
DHARWAD	-7.5	2	1.4	-4.8	11.1	-11.7		6.3	-2.7	-3.3
GADAG	3.2	3.1	9.6	11.1	23.1	-6.5		1.6	-1.8	-2.4
GULBARGA	9.5	-13.8	5.8	17.2	-0.5	2.4			5.1	-1.9
HASSAN	-6	-5.1	-4	-8.3	8	-3.7	-16.4			8.2

HAVERI	3.9	5.6	7.1	5.7	12.5	8.2	-10.6		2.1	2.3	
KODAGU	-6.9	-13.2	-15.5	13.4	-31	-8.6				-15.9	
KOLAR	3	7	5.6	-11.8	17.6	-8.9	-11.1			18.3	
MANDYA	2.5	5.6	7.7	11.1	16.7	10				16.3	
MYSORE	2.4	3.6	10	10	11.8	3.3				13.2	
RAICHUR	-9.2	1	1.4	9.9	-15	10.7		8.5	-15	3.2	
SHIMOGA	4.2	5.4	-13.9	-13.3	20.7	-10.1			-12	4.6	
TUMKUR	-3.9	3.6	-4.4	-18.4	-21	28.4		-13.8		2.4	
UDUPI	4.4	1.6	-19.3	-6.7	-36	-2		-2.8		-7.6	
Uttara Kannada	0.8	-4.7	-16.7	-11.9	-47	-14.1				-13.5	
Net Changes	-0.3	-0.9	1.2	2.6	1.3	1.3	-13.9	-2.8	-1.2	1.19	- 1.1508

- **Table 2. Majority of the area falling under each of the districts are vulnerable for low productivity of the crops.**

Crops	Gulbarga, DK, Hassan, Kodagu and majaority of Northern districts
Rice (Irrigated)	Belgaum, Bellary, Raichur, Kodagu, Dharwad, Chamarajanagar, DK & Hassan,
Aerobic Rice	Belgaum, Bellary, Chikmagalure, DK, UK, Gulbarga and Kodagu,
Maize	Bijapur, Chikmagalure, DK, UK, Udupi, Kodagu, Shimoga and Tumkur
Sorgam	Chikmagalure, DK, UK, Udupi, Davanagere, Hassan, Kolar, Shimoga and Tumkur,
Redgram	Bagalakote, Bellary, DK, Udupi, UK, Kodagu, Raichur and Tumkur.
Cotton	Bellary, Chikmagalure, Chitradurga, DK, UK, Shimoga, Kolar, Kodagu, Dharwad and Gadag.
Potato	Bangalore, Chikmagalure, Davanagere, Hassan, Haveri and Kolar
Soybean	Bangalore, Tumkur and Chitradurga
Ragi	UK, DK, Udupi and Kodagu
Wheat	Raichur and Shimoga

- **<u>Mitigation Strategy:</u>**

To mitigate the climate change effects on agricultural production and productivity a range of adaptive strategies need to be considered. Changing cropping calendars and pattern will be the immediate best available option with available crop varieties to mitigate the climate change impact (Ref. 5). The options like introducing new cropping sequences, late or early maturing crop varieties depending on the available growing season, conserving soil moisture through appropriate tillage practices and efficient water harvesting techniques are also important. Developing heat and drought tolerant crop varieties by utilizing genetic resources that may be better adapted to new climatic and atmospheric conditions should be the long-term strategy. Genetic manipulation may also help to exploit the beneficial effects of increased CO_2 on crop growth and water use (Ref. 6). One of the promising approaches would be gene pyramiding to enhance the adaptation capacity of plants of plants to climatic change inputs (Ref. 7). There is thus an urgent need to address the climate change and variability issues holistically through improving the natural resource base, diversifying cropping systems, adapting farming systems approach, strengthening of extension system and institutional support. Latest improvements in biotechnology and information technologies need to be used for better agricultural planning and weather based management to enhance the agricultural productivity of the country and meet the future challenges of climate change in the dryland regions of the world. Some of the most important mitigation strategies and zone wise adaptations are given in tables 3 & 4 respectively.

Table 3. Mitigation and adaptation for increasing productivity under changed climate scenario.

Crop	**Action Plan**	
	Short-term	**Long-term**
Maize	-Adoption of crop rotation and in-situ incorporation of maize stalks. -Popularization and demonstration of organic manure production method. -In-situ stalk incorporation for manuring. -Popularization of existing hybrid Nithyashree through large-scale demo.	Evaluation of efficient short-duration legumes for in-situ mulching/grain and green manures as intercrops. Development and evaluation

	-Adequate supply of seeds of Nithyashree through farmer's participation seed production. -Seed treatment for control of downy mildew. - Spray Mancozeb for control of Turcicum Leaf Blight and Polysora rust diseases at 35. 45 and 55th day after sowing.	of hybrids for biotic stress.
Ragi	-Popularization of existing varities through large scale demonstration. -Practicing of staggered nursery for timely transplanting. -Promotion of seed treatment with biofertilizers. -Promotion of organic manure promotion structures. -Conducting demonstration or harvesting at RSK levels. -Popularization of existing seed drill and large scale demonstration -Supply of seed-drill on custom hire basis.	-Selection and evaluation from existing genotypes on participatory approach.
Sorghum (Jowar)	-Organizing large scale demonstrations on integrated crop management practices involving moisture conservation, Integrated Nutrient Management (INM) and improved varieties in the cultivators' field. -Increased supply of quality seeds through seed corporations and private seed industries. -Production and supply of seeds through farmers participatory seed production programmes. -Popularization of "seed village" programme through Gram Panchayat or farmer's societies.	Development of early maturity seed varieties with Maldandi (M35-1) yielding ability for shallow soils.
Sunflower	-Screening of germplasm for identification of inbred lines resistant to powdery mildew. -Application of bio-fertilizer for seed treatment. -Crop rotation with legumes. -Intercropping with ground nut/pigeon pea/soyabean. -Use of micronutrients like sulphur and and boron. -Demonstration on 50 ha at each RSK. -Promotion of organic manure production structures and recycling	-Development of new powdery mildew tolerant hybrids through MAS in comparison with powdery mildew tolerant KBSH-53 sunflower hybrid.

	of sunflower crop residues. -Use of 5 kg/ha seed rate with spacing of 60 cms ×30 cms. -Thinning of excess seedlings after 10-12 DAS by retaining one healthy seedling/hill. -Popularization of wider row spacing for mitigating drought. -Demonstration at RSK levels. -Promotion of seed production on participatory approach. -Large scale demonstration for popularization of university hybrids. -Use of need based IPM measures. -Use of right pesticide at proper time against targeted pests. -Conducting demonstrations at RSK's. -Seed treatment and timely management practices for the control of measure pests and diseases.	
Sugarcane	**(For both Short and Long Term)** Evaluation of existing genotypes for all reason planting Promotion of seed production technical know-how and supply of foundation seeds through sugar factories. -Promotion of organic manure production structure, *in-situ* green manure growing. Trash recycling etc. -Encourage farmers to grow legumes as intercrop. -Imparting training to extension personnel regarding balanced use of fertilizers -Verification and validation of existing fertilizer schedule. -Conducting large scale demonstrations in 100 ha. -Demonstration on *in-situ* trash mulching and use of trash for vermicomposting. -Imparting training to extension.	
Groundnut	-Seed treatment with rhizobium. -Promotion of organic manure production structures for residue recycling. -Application of Recommended fertilizer dose and micronutrients and gypsum. -Adoption of crop rotation and intercropping with redgram based system.	Development/evaluation of varieties from existing lines through farmers' participatory approach. -Adoption of effective moisture conservation practices to

	-Conducting demonstrations at each RSKs. -Use of improved seed-cum-fertilizer drill with recommended seed rate. -Use of quality seed. -Supply of seed-cum-fertilizer drill on customer hire basis. -Creating awareness about use of right PP chemical at right time. -Conducting demonstration and training regarding pest management. -Popularize of existing groundnut digger and pod pluckers. Promotion of quality seed production through seed village concept.	mitigate terminal drought. Modification and development of groundnut digger and pod plucker.

Table 4. Adoption strategies for improvement of the productivity:

Name of the Zone	Suggested Cropping Patterns for Different Agro climatic Zones
North–Eastern Transition zone	Area under rained Maize, Sorghum, Cotton and Wheat is proposed to be increased substituting the area under Redgram.
North-Eastern Dry zone	Area under rained Maize, Sorghum and Wheat is proposed to be increased substituting the area under Redgram and other crops.
Northern Dry Zone	The area under rainfed cotton, wheat and sorghum which have shoen higher productivity is proposed to be increased in Bijapur district. Maize, Sorghum and Redgram can be taken more in Gadag district reduce cotton area.
Central Dry zone	The area under Sorghum can be extended to entire Bellary district substituting the area under rice. Rice, Maize, Sorghum, Red gram and Ragi area is to be increased by intercropping sorghum and pearl millet. Cotton and Soya area area is proposed to be restricted to irrigated areas.
Eastern Dry zone	Maize, Sorghum, Redgram and Cotton area is proposed to be increased is substituting a part of rainfed finger millet in Bangalore district and in parts of Tumkur district. In Kolar district and Maize, Redgram and Ragi area can be increased in liew of Sorghum. With the increased rainfall, Mulbery and

	vegetable area can be enhanced during kharif.
Southern Dry zone	Rice area proposed to be reduced by Maize, Sorghum and Redgram crops in the zone. Rice can be increased in Mysore and Mandya district and parts of Hassan district in view of the high yield potential and water availability. Ragi can be continued in the same area.By increasing the cropping intensity under irrigation the area under rice and Sugarcane proposed to be increased.
Southern Transition zone	The area under kharif sorghum proposed to be increased in view of their high productivity. Summer rice area is proposed to be diverted to irrigated groundnut.
Northern transition zone	Because of the high yield potential, area under cotton, wheat, kharif sorghum and groundnut is to be increased. Green gram is to be introduced on large scale as a catch crop in kharif. A part of the area under rainfed rice is to be substituted by maize.
Hilly	Area under Redgram in Chikmagalur is to be increased. Rice area in Shimoga can be enhanced. Apart of the rice in the upland is to be substituted by finger millet. Green gram and black gram are to be introduced in rice fallows. Plantation crops are to be increased.
Coastal	The area under groundnut is to be increased in rice fallows on residual moisture/supplemental irrigation. Black gram is proposed as second crop after rice on residual moisture. In view of the sugar factories, area under sugarcane is to be increased to some extent.

Future work for adaptation and mitigation of climate change in Karnataka:

Altered agronomy of crops: Small changes in climatic parameters can often be managed reasonably well by altering dates of planting, spacing and input management. Alternate crops or cultivars more adapted to changed environment can further ease the pressure.

Development of resource conserving technologies: Recent researches have shown that surface seeding or zero-tillage establishment of upland crops after rice gives similar yields to when planted under normal conventional tillage over a diverse set of soil conditions. This reduces costs of production, allows earlier planting and thus higher yields, results in less weed growth, reduces the use of natural resources such as fuel and steel for tractor parts, and shows improvements in efficiency of water and fertilizers

Improved risk management though early warning system and crop insurance: The increasing probability of floods and droughts and other uncertainties in climate may seriously increase the vulnerability in some area. Policies that encourage crop insurance can provide protection to the farmers in the event their farm production is reduced due to natural calamities. In view of these climatic changes and the uncertainties in future agricultural technologies and trade scenarios, it will be very useful to have an early warning system of environmental changes and their spatial and temporal magnitude. Such a system could help in determining the potential food insecure areas and communities given the type of risk.

Cloud seeding: Cloud seeding can be taken up during prolonged drought situation and also at some critical crop growth stages to protect the crop growth and to finally realize the good productivity. On the assessment of water shortage during drought years (particularly in crop growing seasons) and enhanced area under high water required crops, cloud seeding may be advocated.

Rainwater management in rainfed areas: Various soil and water conservation measures relevant for rainfed agriculture include: *In-situ* measures for rainwater management in rainfall areas, Off season land treatment, *Conservation furrows* , *Ridges and furrows system in cotton, Cover cropping, Micro catchments for tree systems.*

Long term measures for rain water management in rainfed areas: Water harvesting, Contour trenching for runoff collection, On-farm reservoirs, Ground water recharge structure (percolation tanks), Recharge through defunct wells.

In addition to the above, some of the general practices for mitigation of climate change effects.

Drought management strategies in Rainfed Agriculture: In the absence of any assured irrigation facility and with ever growing changing patterns of temperature and precipitation, the rain water management technologies would play a greater role in rainfed areas. Renewed focus with incentive measures for in-situ especially in low to medium rainfall regions with farmers themselves taking the leadership in conservation should be the focal theme. The predominant interventions to overcome climate related impacts in rainfed areas include

- Soil and water conservation practices
- Agronomic interventions
- Nutrient management practices
- Livestock based interventions
- Development of alternate land use plans

These interventions have a role to play in all agro-eco systems except that their order of priority changes, which basically depends on rainfall, status of natural resources like soil, water etc.

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Chapter 5: Impact of Climate Change on Water resources in Karnataka

Executive summary: In this chapter, the vulnerability of water sector to climate change in Karnataka is analyzed. We have focused our investigation on runoff changes in the two major basins in the state, Cauvery and Krishna. The main findings are:

1. Overall, it is projected that river runoff in much of the state for all seasons will decline.
2. Severe drought conditions have been predicted for Krishna basin by 2050 with significantly reduced annual average precipitation (-6%) and runoff (-12%).
3. Though rainfall increases by 2.7% in the Cauvery basin by 2050, runoff decreases by 2 % because evapotranspiration is expected to increase by about 7.5% due to warmer temperatures.

1. Introduction

Impact of Climate change on water resources

The impacts of climate change on water resources have been highlighted in the Fourth assessment report of the Intergovernmental Panel on Climate Change (IPCC) indicating an intensification of the global hydrological cycle affecting both the ground water and surface water supply. The IPCC report also concluded that it is highly likely "**The negative impacts of climate change on freshwater systems outweigh its benefits",** with runoff declining in most streams and rivers. Also different catchments are likely to respond differently to the same change in climate drivers, depending largely on catchment physiogeographical and hydro geological characteristics and the amount of lake or groundwater storage in the catchment. The IPCC has also predicted changes in the total amount of precipitation, its frequency and intensity which on the surplus side may lead to flood like situations in some parts while on the other may create drought like situations if they are on the deficit side. The impacts of climate change are also dependent on the baseline condition of the water supply system and the ability of water resource managers, to respond to climate change in addition to pressures due to increase in demand due to population growth, technology, and economic, social and legislative conditions (Gosain et al., 2006).

a. Surface water resources in Karnataka state

There are seven river systems in the state namely Krishna, Cauvery, Godavari, North Pennar, South Pennar and Palar (Figure 1). The annual average yield in the seven river basins is estimated to be 98,406 Million cubic meters (MCM). However economically available utilizable water for irrigation is estimated at 1695 TMC (State of the Environment Report, 2003). According to the Karnataka state Environment Report (2003), there are 36,679 tanks in the state having a command of 6, 84,518 ha with a total irrigation potential of minor irrigation surface tanks estimated at about 1.0 million hectares.

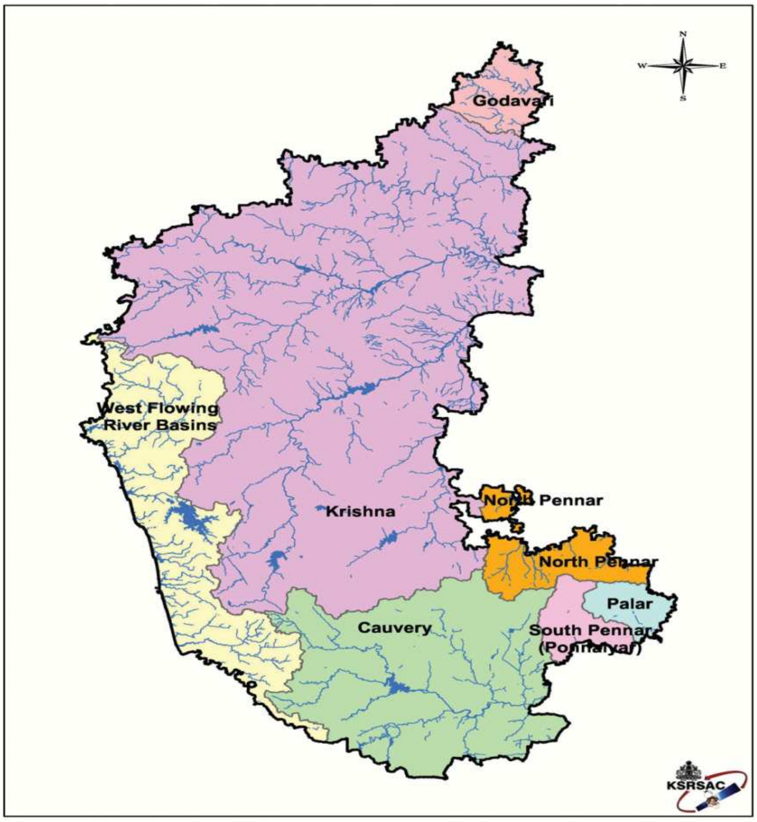

Figure 1: River systems of Karnataka (Source: State of Environment Report of Karnataka, 2003)

2. Objectives

Studies on the impacts of climate change on river basins in Karnataka are limited to national level assessments. A comprehensive assessment of the impacts on river basins in the state has not been attempted. **IISc has initiated a study on an assessment of impacts of climate change on two river basins: Krishna and Cauvery, using a process based model SWAT**. Also, an exercise to study the impact on water resources using the HadCM3 GCM was initiated. This study will also address:

- Modeling of climate change impacts under A1B scenario for 2030s at the district level
- Ranking of districts based on their vulnerability to droughts and floods.

In this report we have presented a synthesis of available information on climate change impacts for two river basins Krishna and Cauvery.

3. <u>River basins selected</u>

Table 1 shows the river systems in Karnataka. It can be observed from the table that the Krishna and Cauvery river basins are the largest in terms of the area drained by them in the state.

Table 1: River systems of Karnataka, their average yield, area drained by them and percentage of water yield (Source: Department of Water resources, Karnataka: http://waterresources.kar.nic.in/index.asp

River System	Estimated average yield (Million cubic meters)	Drainage area (1000 km^2)	Percentage of average yield
Krishna	27,451	113.29	27.90
Cauvery	12,304	34.27	12.33
Godavari	1415	4.1	1.44
West flowing rivers	56,600		57.51
North Pennar	906	6.94	0.92
South Pennar		4.37	
Palar		24.25	
Total	**98,406**	**190.50**	**100**

Krishna basin

With a catchment area of 258,948km^2, Krishna river basin is the fourth largest in India in terms of annual discharge and fifth in terms of surface area covering parts of three states in South India (Figure 2). The principal tributaries of Krishna in Karnataka are Ghataprabha, Malaprabha, Bhima and Tungabhadra. Constructions of large dams on the river for generation of hydroelectricity have severely reduced the river discharge.

Figure 2: Map of Krishna basin (Source: Hydrological Information System NATCOM, 2003)

In a study done by Bower et al. (2006) the impact of climate variability and increasing water consumption on the runoff of the Krishna River was assessed by simulating runoff under both conditions using STREAM water balance model and it was observed that runoff under climate variability varied slightly (6-15%) while due to the construction of the reservoir on river in 1960, the runoff decreased by 61%.

Cauvery basin

The Cauvery river basin (Figure 3) is the second major river system in Karnataka with a catchment area of 81,155 $km^{2.}$. The principal tributaries of Cauvery in Karnataka are the Harangi, the Hemavathy, the Lakshmanathirtha, the Kabini, the Shimsha, the Arkavathi and the Suvarnavathy. All these rivers except the Kabini, Arkavathy and Suvarnavathy Rivers rise and flow fully in Karnataka.

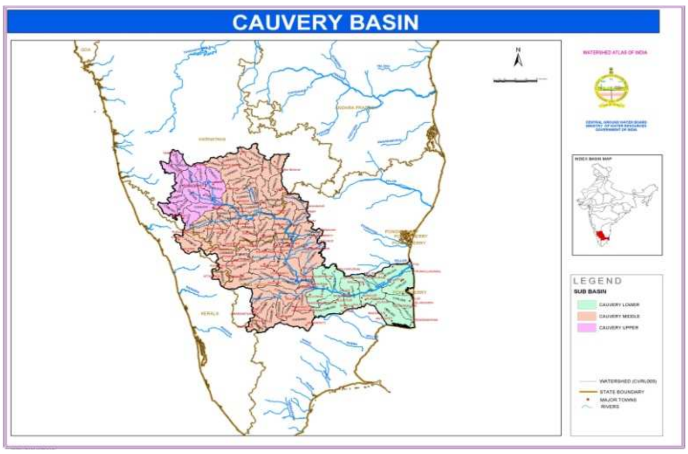

Figure 3: Map of Cauvery basin (Source Source: Hydrological Information System NATCOM, 2003)

a. Methodology

An assessment of the impacts of climate change on water resources can be best handled through modeling of hydrological conditions in river basins under future predicted climate variables like precipitation and temperature. The main components of the hyrdrological cycle are precipitation, evaporation and transpiration. Changes in climate parameters, solar radiation, wind, temperature, humidity and cloudiness affect evaporation and transpiration. The river basin is generally the most appropriate primary exposure unit for assessing impacts on hydrological resources.

Soil and Water Assessment Tool (SWAT): The Soil and Water Assessment Tool (SWAT) is a process-based, spatially distributed, physically based, continuous daily time-step hydrological model which evaluates land management decisions in large ungauged rural watersheds. It is designed to predict long-term nonpoint source pollution impacts on water quality such as sediment, nutrient and pesticide loads. Model inputs include the physical characteristics of the watershed and its sub basins such as precipitation, temperature, soil type, land slope and slope length, width and slope, Hydrologic processes simulated by the model include evapo-transpiration (ET), infiltration, percolation losses, surface runoff, and lateral shallow aquifer and deep aquifer flow. Model outputs include sub-basin and watershed values for surface flow, ground water and lateral flow, crop yields, and sediment, nutrient and pesticide yields.

Data: Spatial data and the source of data used for the study area include:

- Temperature and Precipitation: IMD and CRU gridded dataset
- Digital Elevation Model: SRTM (90 m resolution)/ASTER
- Drainage Network: Digital Chart of the World, 1992
- Soil maps and associated soil characteristics: FAO Global soil
- Land use: Global land use

  Climate scenario: **IPCC A1B scenario.** The Hydro-Meteorological data pertaining to the river basins is required for modeling the catchment. They include daily rainfall; maximum and minimum temperature, solar radiation, relative humidity and wind speed will be generated using the weather generator module of SWAT.

b. Impact of Climate Change

A comprehensive assessment of impact of climate change on hydrology of the river basins was made by Gosain et al. (2006) where, all major river basins of India, except Indus and Brahmaputra, were modeled for future impacts under a GHG scenario using the SWAT model. The impacts were found to be different in different catchments. Table 2 below shows the change in water balance components as a percentage of rainfall for Krishna and Cauvery river basins. SWAT model run for the Krishna and

Cauvery basins under the NATCOM project was done separately using daily weather generated using the weather generated by the HadRM2 control climate scenario (1981-2000) and under the future GHG scenario (2041-2060).

Table 2: Comparison of change in Water balance Components as a percentage of rainfall (Source: NATCOM, 2004)

Basins	**Scenario**	**Rainfall (mm)**	**Run-off (mm)**	**As a proportion of rainfall (%)**	**Actual ET**	**As a proportion of Rainfall (%)**
Cauvery	Control	1309.0	661.2	50.5	601.6	46.0
	GHG scenario	1344.0	650.4	48.4	646.8	48.1
Krishna	Control	1013.0	393.6	38.9	585.0	57.7
	GHG	954.4	346.9	36.4	575.6	60.3

Impact of Climate Change on Krishna basin

Severe drought conditions have been predicted for Krishna basin under the projected climate change scenario, with significantly reduced annual average precipitation, actual evapo-transpiration and water yield in future (Gosain et al., 2006). Using the SWAT derived soil moisture index (SMI), the study has estimated the number of drought weeks to considerably increase during GHG scenario for most sub-basins in Krishna, barring five sub-basins with increasingly constant water stressed conditions to be experienced. Figure 4 below shows the trends in water balance for the Krishna basin under HadRM2 IPCC IS92 control and GHG scenarios.

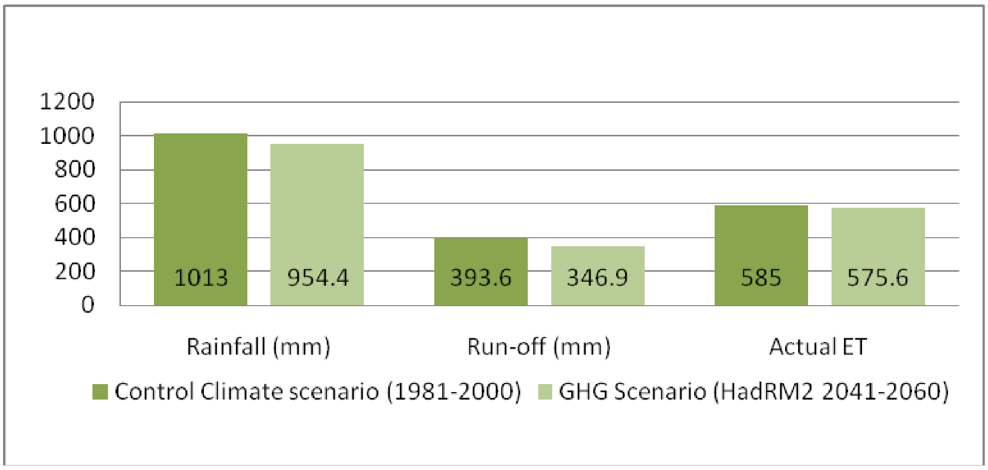

Figure 4: Trends in water balance for the Krishna basin under control and GHG scenarios (Source: NATCOM, 2004)

Impact of Climate Change on Cauvery basin

Figure 5 shows the trends in water balance for the Cauvery basin. It can be observed from Table 2, an increase in rainfall by 2.7% is projected under the Climate change scenario. The increase in rainfall due to climate change does not result in an increase in runoff as may be generally predicted; it results in a reduction in runoff by about 2% while the actual evapo-transpiration is expected to increase by about 7.5%.

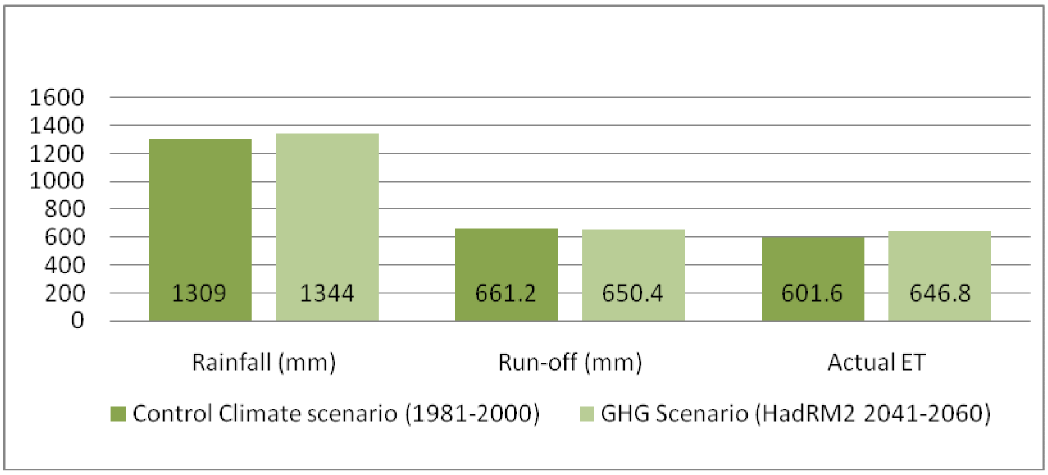

Figure 5: Trends in water balance for Cauvery river basin (Source: NATCOM, 2004)

Impact of climate change on the water sector: HadCM3 GCM: The GCM gives several outputs that are critical to assess the impact of climate change such as total runoff, evapo-transpiration and soil moisture content. In this report, we describe the changes in total runoff due to climate change, aggregated over various districts (Fig. 6). Our salient findings are:

- Overall, there is a projected decrease in runoff for much of the state, for all the seasons
- During the monsoon season, there is a significant decrease in runoff in the North-west parts of the state

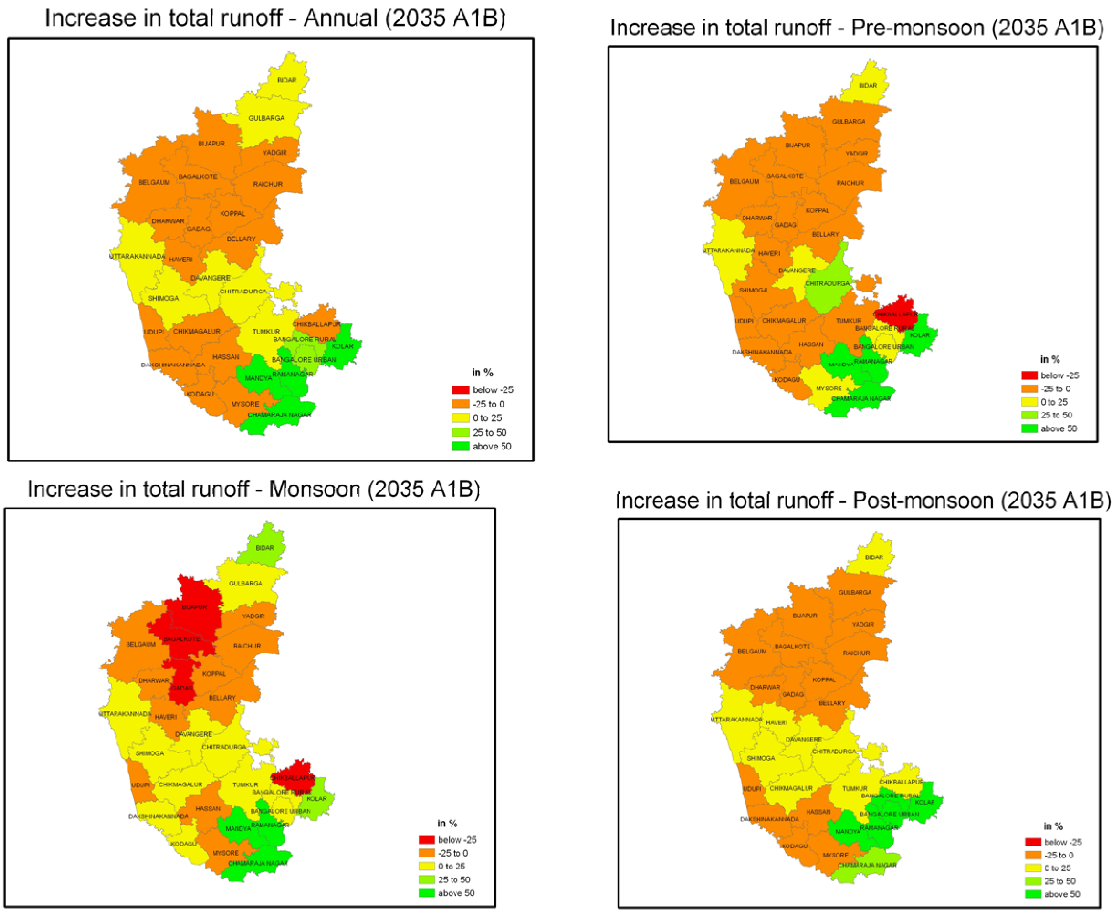

Figure 6: Percent increase in total runoff by 2021-2050 (A1B), with respect to the baseline. The red colour shows significant decrease, while the green color shows increase.

Conclusions:

New A1B scenario climate projections and SWAT model outputs will soon be available for Karnataka at district level for the two river basins. Climate change impact assessment made so far indicate an increased moisture stress in Krishna basin and marginal impact on Cauvery basin. Significant changes in runoff are also projected.

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Chapter 6: Socio-economic Vulnerability and Adaptive Capacity Assessment: An Analysis

1. Background

National Communication of Government of India to UNFCCC has consolidated some of the potential implications arising out of climate change, such as a) increased water stress, b) higher incidences of extreme weather events such as floods and extended droughts, c) rise in sea level and resultant problems to heavily populated coastal belts, d) shifting of forest types and loss of biodiversity, in turn affecting ethnic tribal populations and other relevant dependent activities, e) changes in agricultural productivity due to alterations in hydrological cycle, f) climate change induced negative impact on health etc. It is, universally agreed that unless coping measures are initiated and implemented, climate change would have disastrous impact on welfare. Further, that, sooner the adaptive measures, better would be the gain from such measures.

2. Brief Profile of Karnataka State

Karnataka state has an area of 1,91,791 sq km, (5.83% of the total geographical area of India) and supports a population of about 5 crores. It has four natural regions – the west coast, the Western Ghats or malnad, the northern *maidan* and the southern *maidan*.

The coastal region, a narrow belt that lies between the Western Ghats and the Arabian Sea, encompasses the districts of Dakshina Kannada, Udupi and Uttara Kannada. This region receives heavy rainfall, in the range of 2,500 mm to 3,000 mm. **The Western Ghats or malnad** includes the districts of Chikmaglur, Hassan, Kodagu, Shimoga and the uplands of Dakshina Kannada, Uttara Kannada, Udupi, Belgaum and Dharwad districts. It receives rainfall in the range of 1,000 mm to 2,500 mm. Much of the dense rain forest area of the state lies in this region. **The southern maidan or plateau** Elevation is between 600 metres and 900 metres above sea level. Rice, sugarcane, ragi, coconut and mulberry are the principal crops. **The northern maidan or plateau** primarily includes the Deccan plateau, with its rich black cotton soil.

The Krishna and its tributaries – the Malaprabha, Ghataprabha, Tungabhadra, Bheema and Karanja – sustain agriculture here. It is a low rainfall area where jowar, cotton, oilseeds and pulses are cultivated. Agro climatically, the state is divided into ten agro climatic zones.

Population Trends

Karnataka has a population of 53 million (2001) accounting for 5.13 per cent of India's population. The sex ratio of 965 in the state stands above the all-India average of 933. The population density in the state is 275 as compared to 324 at the all-India level in 2001. About 66 per cent of the population in the state lives in rural areas. The Scheduled Caste population constitutes about 16.2 per cent of the total population in Karnataka.

Per Capita Income

From the data available with Directorate of Economics and Statistics, per capita income levels for districts were computed across the years. As expected, Bangalore district has occupied top position and Bidar District is at lowest position.

Sector wise district income levels: In all districts, contribution from primary sector is decreasing from 1990 -91 to 2001-02, indicating prima facie, shift from unskilled to skilled employment, migration, increased role of tertiary sector. Similar trend is observed in the case of secondary sector as well. Interestingly, Net District Domestic Product (NDDP) over years shows constant increase in all sectors.

Analyzing the same from climate change point of view, it becomes crystal clear that primary sector remains very important for overall welfare of State and this point becomes critical if one were to look into occupational trends.

Agricultural Activities and Irrigation

Cultivators and agricultural laborers form about 56 per cent of the workforce (2001 census). Further, extent of arid land in Karnataka being second only to Rajasthan in the country, agriculture is highly dependant on the vagaries of the southwest monsoon. Out of the net area sown, only 25 per cent is irrigated. Similarly, Hasan district supports highest percentage of agricultural labour across the districts followed by Chamraj Nagar district. In terms of irrigation, it is Shimoga district which has highest percentage of irrigated area followed by Mandya. In terms of livestock, Gulburga registered highest.

Occurrence of Drought

With reference to drought prone area in the state, Bijapur district has a record of 73% of its area as drought prone and it is closely followed by Chitradurga at 70% of its area (Table 1).

Table 1: Identified Drought Prone District of Karnataka (2002)

Year 2002	No. of Talukas	Dist area Sq Km	Taluka affected by drought	Area affected by drought Sq Km	% area affected
Karnataka/Dist	139	152163	42	57646	38
Bangalore	11	7949.5	-	-	-
Belgaum	10	13460.8	1	1996	15
Bellary	8	9548.5	3	3994.3	42
Bijapur	11	17092.8	7	12477.4	73
Chikmangalur	7	7222	1	804.8	11
Chitradurga	9	10754.5	5	7477.5	70
Dharwar	17	13480.1	3	2772.32	21
Gulbarga	10	16167.8	5	8131	50
Hasan	8	6833.3	1	1277.8	19
Kolar	11	8215.2	4	3444.7	42
Mandya	7	4961	1	1034.28	21
Mysore	11	11947	1	1235.9	10
Raichur	9	13972.4	4	6347.6	45
Tumkur	10	10557.7	6	6651.9	6

Source: Compendium of Environment Statistics, 2002, Central Statistical Organization, Ministry of Statistics and Programme Implementation, GOI

From the above data, following interpretations can be drawn,

- Primary sector is very important economic activity and has wide diversification from rain-fed agriculture to livestock rearing to marine fishing and is livelihood support from majority of population in the state.
- As outlined in National communication, GoI, climate change, among others, would affect primary sector severely.
- To minimize such negative impacts, suitable interventions are required.

For identifying proper intervention and districts to be implemented, as mentioned earlier, livelihood support base was calculated on the basis of working population involved in various sectors. For this purpose, economic activities are divided into sectors of different orders depending on the extent of influence of climate change. For instance, rainfed agriculture is directly influenced by the amount and time of the rainfall, on the other hand, education sector influenced to lesser extent. Thus, it is possible to group various economic sectors on the basis of the extent of influence of climate has on the performance of these sectors. On such grouping is given below:

a) First Order

b) Second Order, and

c) Third Order sectors

First Order sectors are those whose performance is directly influenced by the climatic variations, such as Agriculture, Fisheries, Forestry, and Health. Second Order sectors are those susceptible to climatic conditions directly and indirectly as well, but certain measures could reduce the influence of climatic variations, for instance Mining, Construction, Household Industries, Education, Transport, communication. Third Order sectors are represented by those sectors which are relatively independent to large extent of climatic variations such as Service Sector, industry and others. Further, it is possible to rank the districts of state on the basis of size of population dependent on that particular sector. This exercise would give us an idea about the percentage of district population vulnerable to climatic conditions. Census data of 1991 and 2001 was used to rank the vulnerability of districts.

1st Order Sectors

Primary economic activities such as agriculture, fishing activities, livestock rearing etc. are directly influenced by the climatic factors and indirectly the size of population dependent on these sectors. Depending on the size of population engaged in 1st Order economic activities, districts were ranked and given in Table 2.

Table 2: Ranking of Districts According to Percentage of Dependent Population

	1991	2001	1991 Census		2001 Census	
District	Total Population	Total Population	1st Order (%)	Rank	1st Order (%)	Rank
Bagalkot	1,652,232	1651892	22.7	17	28.3	16
Bangalore	6,523,110	6537124	2.6	27	2.3	27
Bangalore(R)	1,877,416	1881514	23.5	15	29.3	13
Belgaum	4,207,264	4214505	22.4	19	30.7	11
Bellary	2,025,242	2027140	25.5	9	30.2	12
Bidar	1,501,374	1502373	23.3	16	23.2	20
Bijapur	1,808,863	1806918	25.1	11	27.9	17
Chamaraja Nagar	964,275	965462	28.1	3	32.9	7
Chikmagalur	1,139,104	1140905	22	20	22.5	22
Chitradurga	1,510,227	1517896	25.7	8	34.2	5
Dakshina Kannada	1,896,403	1897730	10	26	4.9	26
Davanagere	1,789,693	1790952	24.3	14	28.6	15
Dharwad	1,603,794	1604253	18	23	22.6	21
Gadag	971,955	971835	25.9	7	32.8	8
Gulbarga	3,124,858	3130922	24.9	12	28.9	14
Hassan	1,721,319	1721669	24.5	13	35	3
Haveri	1,437,860	1439116	27.2	4	34.4	4

Kodagu	545,322	548561	14.1	25	5.9	25
Kolar	2,523,406	2536069	25.4	10	31.2	10
Koppal	1,193,496	1196089	28.1	2	33.7	6
Mandya	1,761,718	1763705	28.2	1	35	2
Mysore	2,624,911	2641027	19.4	21	24.5	19
Raichur	1,648,212	1669762	26.7	6	32.1	9
Shimoga	1,639,595	1642545	22.6	18	26.9	18
Tumkur	2,579,516	2584711	26.9	5	35.4	1
Udupi	1,109,494	1112243	18.8	22	16.6	24
Uttara Kannada	1,353,299	1353644	16.5	24	16.8	23
State Average	52,733,958		20.7		24.8	

Details of ranks (given on the basis of percentage of population in select economic activities) show that the place of districts varied from 1991 to 2001. For instance, Tumkur which was having 5th position in terms of population in involved in 1st order activities has moved upwards to rank 1. In other words, size of population involved in first order activities has increased from 1991 to 2001. Similarly, Koppal district which was in second position as per 1991 Census has less population in 1st order sector in 2001 Census. To figure out probably reasons, in-depth studies are required to be undertaken.

2nd Order Sector

Economic activities such as construction, household activities are influenced by climate change, but impact are not severe as compared to the 1st Order activities. An effort was made to rank the districts on the basis of population involved in 2nd order activities. Details are given in Table 3.

Table 3: Ranking of Districts Dependent on the Population in 2nd Order Activities

	1991	2001	1991 Census		2001 Census	
District	Total Population	Total Population	2nd Order	Rank	2nd Order	Rank
Bagalkot	1,652,232	1651892	2.1	2	3.3	3
Bangalore	6,523,110	6537124	2.6	1	1.1	19
Bangalore(R)	1,877,416	1881514	1.6	14	2.3	5
Belgaum	4,207,264	4214505	1.7	7	1.6	12
Bellary	2,025,242	2027140	1.9	4	1.3	14
Bidar	1,501,374	1502373	1	25	0.9	23
Bijapur	1,808,863	1806918	1.2	23	1.2	16
Chamaraja Nagar	964,275	965462	1.6	11	2	6
Chikmagalur	1,139,104	1140905	1.3	20	1.1	20
Chitradurga	1,510,227	1517896	1.6	9	1.6	10
Dakshina Kannada	1,896,403	1897730	1.7	8	10.6	1
Davanagere	1,789,693	1790952	1.3	18	1.7	9
Dharwad	1,603,794	1604253	1.6	10	1.2	15
Gadag	971,955	971835	1.7	6	1.7	8
Gulbarga	3,124,858	3130922	1.6	13	1.1	18
Hassan	1,721,319	1721669	1	27	0.8	26
Haveri	1,437,860	1439116	1.2	22	1.9	7
Kodagu	545,322	548561	1.3	21	0.5	27
Kolar	2,523,406	2536069	1.5	16	1.6	11

Koppal	1,193,496	1196089	1.3	19	1.5	13
Mandya	1,761,718	1763705	1.2	24	1	21
Mysore	2,624,911	2641027	1.5	15	0.8	25
Raichur	1,648,212	1669762	1	26	0.9	24
Shimoga	1,639,595	1642545	1.4	17	1.1	17
Tumkur	2,579,516	2584711	1.6	12	2.4	4
Udupi	1,109,494	1112243	1.7	5	5.6	2
Uttara Kannada	1,353,299	1353644	2	3	1	22
State Average	52,733,958		1.6		1.8	

As in the case of 1st Order ranking, 2nd order ranks of districts are also varying over the time. Similarly, districts were ranked according to the 3rd Order activities and details are given in Table 4.

Table 4: Ranking of Districts Depending on Population in 3rd Order Sectors

	1991	2001	1991 Census		2001 Census	
District	Total Population	Total Population	3rd Order	Rank	3rd Order	Rank
Bagalkot	1,652,232	1651892	6.9	14	12	20
Bangalore	6,523,110	6537124	19.6	2	35.9	2
Bangalore(R)	1,877,416	1881514	7.1	13	15.9	10
Belgaum	4,207,264	4214505	7.3	11	12.3	19
Bellary	2,025,242	2027140	7.1	12	13.9	13
Bidar	1,501,374	1502373	6.4	17	13	17

Bijapur	1,808,863	1806918	5.1	26	10.7	26
Chamaraja Nagar	964,275	965462	6.1	21	11.5	23
Chikmagalur	1,139,104	1140905	6.6	16	21.7	5
Chitradurga	1,510,227	1517896	6	22	11.8	21
Dakshina Kannada	1,896,403	1897730	22.7	1	34.5	3
Davanagere	1,789,693	1790952	7.7	9	13.5	14
Dharwad	1,603,794	1604253	11.2	4	18.8	7
Gadag	971,955	971835	7.7	10	12.6	18
Gulbarga	3,124,858	3130922	6.1	20	13.1	16
Hassan	1,721,319	1721669	6.2	19	14.4	12
Haveri	1,437,860	1439116	5.9	23	10	27
Kodagu	545,322	548561	9.3	6	42.2	1
Kolar	2,523,406	2536069	6.8	15	15.9	9
Koppal	1,193,496	1196089	5	27	11.1	24
Mandya	1,761,718	1763705	5.7	24	11.7	22
Mysore	2,624,911	2641027	9.6	5	16.7	8
Raichur	1,648,212	1669762	5.6	25	11	25
Shimoga	1,639,595	1642545	8.3	8	15.5	11
Tumkur	2,579,516	2584711	6.2	18	13.1	15
Udupi	1,109,494	1112243	14.2	3	21.7	6
Uttara Kannada	1,353,299	1353644	8.9	7	25.1	4
State Average	52,733,958		9.3		17.9	

Observations

- Climate Change will have its negative impact primarily on the first order economic activities and Tumkur, Mandya, Hasan and Haveri with population percentages of 35.4, 35.2, 35, and 34.4 respectively would be negatively affected. On the other hand, Udupi, Kodagu, Dakishna Kanada and Bangalore districts are affected least as they have lowest percent of population dependent on the first order activities.

- With reference to Second Order Activities, it is again, Tumkur and Mandya districts are susceptible to climate change as they have 35.4% and 35% of working population is dependent on these sectors.

- Adaptation measures/ plans, thus, should focus on Tumkur, Mandya, Hasan, Haveri districts with focus on alternate/supplement activities in these districts.

3. Climate Change Vulnerability

Vulnerability as a social response (measure of resilience to hazard), and place based vulnerability that combines both exposure and societal response. The socio-economic Vulnerability focuses on the internal state of the system that makes human societies and communities either susceptible to or cope with damages form external hazards (Brook, 2003). In short, internal state of the system is determined independently with existing social, economic, political and environmental factors, and hence considered hazard as the exogenous variables (Adger and Kelly, 1999; Brook, 2003). In the Indian context, this notion was extensively studied in the case of agriculture, which is highly sensitive to the climate change. Kumar and Parikh (2001), following the Ricardian approach, reported that the loss of farm level net revenue was around 8.4 per cent under the given climate scenario.

At state level studies, High Power Committee (HPC) for Redressal of Regional Imbalances under the Chairmanship of Prof. D.M. Nanjundappa has considered 35 different indicators for classifying all the talukas and districts of Karnataka into backward, more backward and most backward. It can be seen from Table 8, that most of the backward talukas in the state are located in northern parts of the state. This indicates appropriate project interventions in the northern Karnataka regions on priority. Further it can be seen from Table-5 that a large proportion of talukas in Karnataka are most backward falling in Gulbarga division of the northern Karnataka.

Table 5: Regional Imbalances in Karnataka

Divisions	Most Backward	More Backward	Backward	Total
Gulbarga	21	5	2	28
Belgaum	5	12	14	31
Bangalore	11	13	9	33
Mysore	2	10	10	22
Total	39	40	35	114

However, HPC Report does consider indicators relevant to 'development", therefore, by considering climate change relevant parameters, vulnerability index at district levels was computed.

Measurement of Vulnerability

Vulnerability is often reflected in the state of the economic system as well as the socio-economic features of the population living in that system. By considering climate change relevant parameters, vulnerability index at district level was computed based on the following dimensions:

1. Demographic and Social
2. Occupational
3. Agricultural
4. Climatic

The index attempts to capture a comprehensive scale of vulnerability by including important indicators that serve as proxies.

Due to unavailability of data for all districts, the present report used data for 27 districts of Karnataka State. We have used data from Karnataka at a glace to calculate the composite vulnerability Index across the districts of Karnataka. Thus, computed Vulnerability Index of districts across the state is given in Table-6.

Table 6: District-wise Vulnerability Indices of Karnataka

District	Index of Demographic and Social Vulnerability	Rank	Index of Occupational Vulnerability	Rank	Index of Agricultural Vulnerability	Rank	Index of Climatic variability	Rank	Composite Index	Rank of Composite Index
Gulbarga	0.5780	2	0.5467	7	0.7169	1	0.7897	2	0.6578	1
Raichur	0.6616	1	0.5928	1	0.5698	7	0.6959	4	0.6300	2
Bijapur	0.4350	12	0.5472	6	0.5174	14	0.9294	1	0.6072	3
Chitradurga	0.4877	7	0.4938	14	0.5219	13	0.6908	5	0.5486	4
Tumkur	0.3580	18	0.4663	18	0.6143	3	0.7150	3	0.5384	5
Bellary	0.5483	4	0.4520	21	0.5122	17	0.6303	6	0.5357	6
Kolar	0.4404	10	0.4959	13	0.5713	6	0.5103	7	0.5045	7
Koppal	0.5567	3	0.5719	4	0.4160	23	0.3333	9	0.4695	8
Bidar	0.5255	5	0.5345	8	0.5164	15	0.2090	15	0.4463	9
Haveri	0.4434	9	0.5275	9	0.5238	12	0.1979	16	0.4232	10
Bagalkot	0.4523	8	0.5065	12	0.4033	24	0.3091	11	0.4178	11
Chikmagalur	0.3732	16	0.3820	23	0.5831	5	0.3284	10	0.4167	12
Mysore	0.4321	13	0.5072	11	0.4603	19	0.2382	14	0.4095	13
Chamarajanagar	0.5089	6	0.5883	2	0.4293	22	0.1111	26	0.4094	14

Uttara kannada	0.2858	24	0.5563	5	0.6492	2	0.1199	25	0.4028	15
Mandya	0.2963	23	0.5849	3	0.4427	20	0.2777	13	0.4004	16
Hassan	0.3324	20	0.4748	17	0.5914	4	0.1751	19	0.3934	17
Davanagere	0.4390	11	0.4917	15	0.5008	18	0.1244	22	0.3890	18
Belgaum	0.4040	14	0.4650	19	0.5276	11	0.1206	24	0.3793	19
Shimoga	0.2993	22	0.4798	16	0.5439	9	0.1869	18	0.3775	20
Bangalore(r)	0.3345	19	0.4561	20	0.5597	8	0.1383	21	0.3722	21
Dharwad	0.3157	21	0.3774	24	0.3880	25	0.4067	8	0.3720	22
Gadag	0.3606	18	0.4374	22	0.3652	27	0.2789	12	0.3605	23
Udupi	0.2592	27	0.5103	10	0.5320	10	0.1207	23	0.3555	24
Bangalore	0.3861	15	0.1694	25	0.3723	26	0.1891	17	0.2793	25
Kodagu	0.2818	25	0.0799	27	0.5156	16	0.1207	23	0.2495	26
Dakshina kannada	0.2654	26	0.1250	26	0.4378	21	0.1553	21	0.2459	27

Table-6 shows the value of the vulnerability index across the districts of Karnataka. In the table rank 1 shows maximum vulnerable district and the vulnerability increase as we go on increasing the rank. In Karnataka, Gulbarga district is the most vulnerable district when we calculate the composite index of a few important indicators such as demographic and social, occupational, agricultural and climatic indicators. According to the composite vulnerability index, Dakhina Kannada is the least vulnerable district of Karnataka.

After developing Composite District Level Vulnerability Index, for identification of suitable interventions at district level and sector wise, dependent population in various economic sectors was considered.

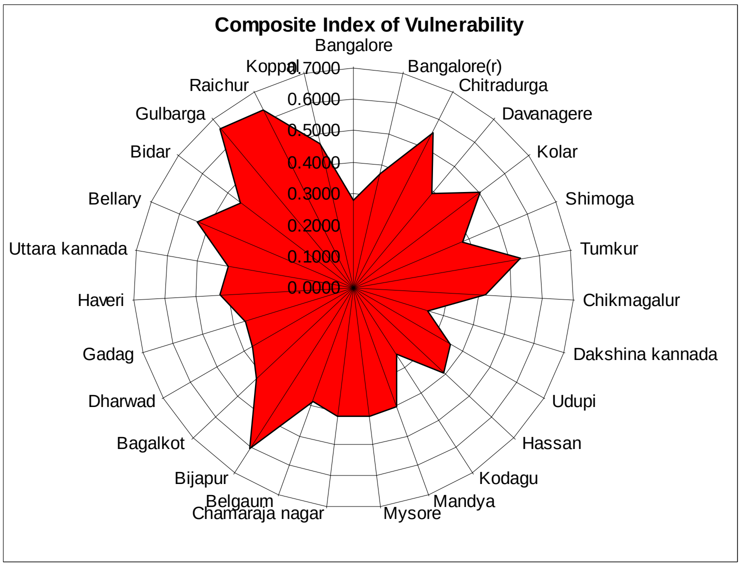

Fig 1: Composite index of Vulnerability across Districts of Karnataka

Chapter 7: Mitigation Options in Energy Sector

Introduction

Today the total annual emissions from Karnataka will be around 80 million tons of CO_2 eq[16]. LULUCF constitutes a negligible amount of net emissions[17] and hence is ignored. In order to project future emissions, one needs to project energy required, future technology developments and the fuel mix. An estimate of elasticity to GDP of that sector is needed to project production or growth of any sector. However, estimating the elasticity to use in such a projection is fraught with difficulty, as future elasticity trends would depend on several macro-economic factors such as relative fuel prices, technology advancement, development priorities of the state, structural shift in the state’s economy, etc. In this section all projections have been undertaken limited up to the year 2020-21 as it is difficult to project technology developments further into the future.

Projecting forward to 2020-21, assuming the gross state domestic product (GSDP) was to grow at 8% annually in line with the country; the total emissions could reach to over 244 million tons of CO_2 eq. in the business as usual case. However, this can be reduced by adopting GHG mitigation strategies. By increased use of energy efficient appliances, adopting energy efficient measures in manufacturing and by introducing greater percentage of energy generated from renewable energy, the GHG emissions level can come down in the future.

India has currently in place policy, regulatory and legislative structure towards GHG mitigation. The integrated energy policy was adopted in 2006. Since then the National Action Plan for Climate Change (NAPCC) was announced in 2008. Two of the core missions announced as part of the NAPCC, are the National Mission for Enhanced Energy Efficiency (NMEEE) and the National Solar Mission (NSM). These plans can help direct Karnataka to reduce the future energy demand and to improve the fuel mix. These policies in turn can help bring down the GHG emissions substantially. Below some of the key technology options that will help mitigate GHG emissions along with the current plans of the government of Karnataka are discussed.

[16] Section on GHG Inventory of Karnataka
[17] Per 2004 NATCOM report this is true for India, and we assume the same for Karnataka

Power Sector

Hydro electricity in 2010 constituted about 27% of the total energy mix in the state, while coal constitutes 58%[18]. For comparison, hydro makes up 15% and coal accounts for 72% of the national generation. However, in determining the future energy mix for the state of Karnataka, one needs to consider resource limitations, cost of generation and other social and political constraints. Going forward the percentage of these renewable sources in the total energy mix is unlikely to increase given the severity of the constraints.

As of January 2010, the total installed capacity of utilities, including allocated shares in joint and central sectors, and captive generation was around 11,959 Mega Watts (MW)[19] as of January 2010. In the period 2009-10, the net generation after considering auxiliary consumption was around 44 billion kWh and the associated CO2 emissions were 28.7 million tons[1]. As of January 2010, the peak power deficit was 22.8% and the energy deficit was 13%[20]. If the generation for this period were 50 billion kWh the state would have been able to meet its electricity demand. If this were projected using an elasticity of 0.95[21] assuming a GSDP growth rate of 8% for Karnataka, the electricity demand required in 2020-21 would be 112 billion kWh in the business as usual case. However, adhering to strict energy efficient programmes the demand could be reduced by 10-15% or to 95 to 100 billion kWh by 2020-21 by adopting DSM options.

Energy Efficiency and Demand Side Management

Promoting energy efficiency (EE) and demand side management (DSM) can help rein in the growth of demand of electricity. This would imply a reduced demand in the future and thus a reduced carbon footprint for the state. Several EE measures are cheaper on life cycle analysis basis and should be considered as means of reducing the demand. The Bureau of Energy Efficiency (BEE) under the NMEEE has several programmes to encourage reduced consumption of energy in buildings, industry and irrigation.

Transmission loss in Karnataka is 4.2% in 2009-10 which is on par with international standards. In 2008-09 the average distribution losses were 20% that includes both technical and commercial losses[22]. In the cities, the technical losses are marginal while they are very high in the rural areas with 11 KV feeder lines running very long distances. Re-engineering of these feeder lines and LT lines in

[18] Ministry of Power, Annual Report 2010
[19] Ministry of Power, Annual Report 2010
[20] Government of Karnataka, Department of Energy
[21] Integrated Energy Policy, 2006
[22] KPTCL

the state would help reduce technical losses. The first step to achieve this would be to carry out the energy audit of the distribution system to segregate and pinpoint the commercial and technical losses. Having identified the causes of technical losses, the DISCOM must analyze all the causes of the technical losses and develop them in the form of several DPRs (detailed project reports). Finally a list of projects showing gestation period, investment, and benefits must be listed so that the projects can be prioritized[23].

The increased use of energy efficient lighting such as compact flouroscent lamps (CFL) and Light Emitting Diodes (LED) has a large potential in bringing down the total consumption of electricity due to lighting in buildings – residential and commercial. Under the 'Belaku scheme' in Karnataka (the national 'Bachat Lamp Yojana') the power utility in exchange for incandescent bulbs will provide 4 CFLs at a subsidize rate of Rs 15 each to each household. Similarly, the use of energy efficient five star labeled appliances such as air-conditioners and refrigerators have an enormous potential to reduce energy consumption. Bangalore Electricity Supply Company's (BESCOM) success of shaving off morning peak load by incentivizing the use of solar water heaters should be replicated in other parts of the state. Public and commercial buildings should be mandated to install solar water heaters.

In the agricultural sector, most irrigation pumps in Karnataka are grid connected and hence diesel based pumps are rare. As grid power is heavily subsidized, the farmers do not have any incentive to switch to efficient pump sets (The majority of the pumpsets in operation operate at an average efficiency of around 30%). Hence most of the irrigation pumps are highly inefficient and are massive consumers of power. Better load management, reducing the water consumption along with replacement of the existing pumps with efficient pumps can result in very large savings in this sector. However, due to political repercussions this is difficult and requires strong political will.

Clean Coal Technologies

Coal will remain the mainstay for electricity generation for the immediate future. Moreover, at present coal is one of the lowest cost options and will continue to be so for a large percentage of total generation by 2020 as well. To meet the projected demand of the state the coal based generation might have to increase considerably in capacity by 2020. Today all the coal based plants in the state of Karnataka as well as in the country are based on sub-critical technology with an average specific CO2

[23] Change Management in Power Distribution, Distribution Reform, Upgrades and Management (DRUM) Training Program, Report prepared by USAID for Ministry of Power

emission of about 1.1 kg per net kWh. However, the new 500 MW sub-critical plants will have a lower specific emission of 0.93 kg per net kWh[24,25].

To reduce the specific emissions from coal based generation, the combustion efficiency can be improved by the use of super-critical technology. The plants based on this technology operate at higher temperature and pressure than the sub-critical ones leading to lower specific emission of 0.83 kg per net kWh. This technology is available and costs almost the same as sub-critical technology. NTPC has plans to set up five 800 MW of supercritical plants in Bijapur district[11]. However, it is not clear how much of this plan will be realized.

Natural Gas generation

Gas based power plants have low capital cost (around Rs 2.7 crores/ MW), even lower than coal based generation, and have a much lower specific emission of 0.42 kg per net kWh and hence an attractive option. However, the availability of gas is a concern. Gas Authority of India Limited (GAIL) is planning a 746 km of gas pipeline from Dhabul in Maharashtra to Bangalore via Belgaum, Gadag Davanagere and Tumkur in Karnataka. This will help setting up gas based power plants along route of the pipeline. There are two combined cycle power plants planned in Bidadi and Tadadi of 1400 MW and 2100 MW[11] respectively.

Hydro Power

Karnataka is endowed with a large hydro potential of about 7,750 MW[26], however, today the installed capacity of large and small hydro together is only around 3,763 MW. Full exploitation of hydel potential is fraught with difficulty due to environmental concerns, people displacement problems and inter-state water disputes. Currently a 400 MW hydel scheme in Hassan district and a 345 MW seasonable scheme at Shivasamudram are awaiting environmental clearance from the Ministry of Environment and Forestry (MoEF). Hence, going forward mini and pico hydel projects are more likely to come up than large projects. Though generation of hydro power is carbon free, the environmental impact of large hydro plants also has to be taken into consideration.

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[24] Discussion with NTPC
[25] *Future of Coal*, Massachusetts Institute of Technology, 2007.
[26] Government of Karnataka, Planning Programme Monitoring and Statistics Department. http://www.planning.kar.nic.in/sites/planning.kar.nic.in/files/AnnualPlan2011-12/vol-I/10%20-%20Chapter10-F_199-218_.pdf

Biomass

Biomass cogeneration in sugar mills is promising in the state, as about 165,000 tons of sugar cane is crushed daily in Karnataka[27]. Currently 623 MW of biomass based power plants are in operation. A large percentage of this is perhaps based on sugar cane bagasse. In addition, the state has accorded permission to 54 new and old sugar factories to establish co-generation and up to 600 MW is expected to be exported to the grid from these after captive usage[9]. Biomass based power is considered carbon neutral and hence no CO2 emissions is attributed due to this.

Solar and Wind

Karnataka Power Corporation Limited (KPCL) has set up three 3 MW solar photovoltaic plants and more are planned making Karnataka an early mover into the utility scale solar PV based power generation. Most parts of Karnataka are reasonably endowed with global radiation suitable for large scale adoption of solar PV. The northern part is said to have sufficient direct normal irradiance (DNI) that is necessary for concentrated solar thermal power (CSP) generation. Given this and the impetus provided by the National Solar Mission, solar energy can play a large role in the power generation of the state. However, the current impediment of solar technology is the high cost of solar technology. Again careful planning, implementing and continuous servicing and maintenance of these plants will lead to successful generation of electricity from these technologies. The state could also consider incentivizing roof top PV systems to abate the usage of diesel in urban buildings.

About 15% of the total capacity and 6% of the total electricity generation in the state is from wind power. With an installed capacity of 1448 MW of wind, Karnataka has the fourth largest installed capacity in the country after Tamil Nadu, Maharashtra and Gujarat. The theoretical installable potential for Karnataka is very high around 8,600 MW[28], however it is unclear how much of this will be realizable.

Nuclear Power

The Nuclear power plant at Kaiga has 4 units of 220 MWs each (Pressurized Heavy Water Reactors technology). Three of these were recently commissioned. There is a proposal to expand the generating capacity by adding one more unit of 500 MW in the near future. Overall nuclear power can contribute

[27] Government of Karnataka, Planning Programme Monitoring and Statistics Department. http://www.planning.kar.nic.in/sites/planning.kar.nic.in/files/AnnualPlan2011-12/vol-I/10%20-%20Chapter10-F_199-218_.pdf
[28] Centre for Wind Energy Technologies; http://www.cwet.tn.nic.in/html/departments_wra.html

up to 1380 MW by 2020 if pursued diligently. Generation of electricity from nuclear power plants is almost carbon free and hence the emission from this is considered zero.

Conclusion

The table below gives the current and possible energy mix in 2020-21, in the business as usual case. It has been assumed that almost all of the planned coal based power plants will be commissioned while others may not be fully realized. As per discussions above, Karnataka has been a forerunner in exploiting several of the low carbon renewable resources and it has a high percentage of energy from renewable in its energy mix. However going forward it is unlikely for the energy mix to have such a high percentage from carbon-free sources due to the constraints discussed above. Perhaps if prices were to come close to grid parity, solar energy may play a role. However, by 2020 it is doubtful if it can contribute more than 2-3% of the generation. Beyond 2020 one can perhaps believe that solar might be able to scale up and efficient coal technologies might order of the day by then. However, coal is likely to dominate the energy mix for the foreseeable future. The CO2 emissions in 2020-21 would be around 84 million tons in the business as usual case from the current level of 29 million tons. The adoption of DSM measures can bring this down by 15% to 71 million tons. This can be further brought down by shift to more carbon friendly sources.

Table 4: Power Sector – Current and Projected

	2010[29]		**2020- Business as usual scenario**	
	Total installed Capacity (MW)	**Net Generation (Billion kWh)**	**Total installed Capacity (MW)**	**Net Generation (Billion kWh)**
Coal Sub-critical	3,903	23.59	12,600	81.24
Coal-supercritical	-	-	400	2.58
Gas	220	0.84	440	1.87
Diesel	333	1.28	460	1.96
Nuclear (PHWR)	195	0.62	880	5.55
Hydro + Small hydro	3,763	11.44	4,200	12.77
Wind	1,448	2.49	2,000	3.43
Biomass + Cogeneration	623	2.03	800	2.61

[29] Ministry of Power, Annual Report 2009-2010

Solar	10	0.02	100	0.16
Captive Generation	1,000	2.01	-	-
Total(Utility + Non Utility)	11,495	44	21,880	112
CO2 emission (million tons)		29		84

Transportation

The transportation sector is a large emitter of CO2 and its share of GHG emissions have been increasing. India imports 80% of its petroleum requirements, and a significant percentage of this is used for transportation. For Karnataka the emissions due to transportation were 8.35 million tons in the year 2007-08. At a state GDP growth rate of 8%, assuming an elasticity of 1, the emissions due to this sector is likely to be 23 million tons.

To lower emissions in this sector, efficient modes and technologies must be used and inefficient ones should be discouraged through policy or fiscal instruments. Some of the following measures will help the state reduce emissions and many of these measures will help improve the overall infrastructure of the state as well.

- Rail freight is considerably more energy and carbon efficient than road freight. The falling trend in the rail freight and the gain in share of the road freight should be reversed.
- Increase the share of public transportation in the cities – increased buses, metro.
- Bicycle lanes should be made available where possible in all the cities.
- Minimum efficiency standard for the country's vehicle fleet should be defined. Fuel efficiency should be improved by imposing periodically tightening fleet efficiency, with mechanism to penalize non-conformance. The state government should work with the BEE to establish these standards.
- The hybrid electric vehicle combines the internal combustion engine (ICE) with an electric propulsion system. The electric vehicle's 'tank to wheel' efficiency is higher by a factor of three than a conventional IC engine vehicle.
- The state government has set up a task force to with the objective to prepare a road map for the GoK to make transition towards Electric/Hybrid public transportation and to develop a few prototypes to run within Bangalore and in a few select inter-city corridors in the state in the upcoming year. If the state were to transition into low-carbon public transportation system, it will be a great achievement and should be pursued.

- Use of bio-fuels could potentially lead to net zero carbon emissions on a net life cycle basis. Recent reports suggest that the state government is considering this as an option. However, we caution large scale thrust to bio-fuels because of the concerns of food security and inflation.

Industries

In the industrial sector, cement, iron & steel, aluminium, textile and paper are some of of energy intensive manufacturing industries. While the large plants are generally very efficient, efficiency often reaching the global best standards, the small and medium scale plants are often less efficient. The government of India's Perform Achieve and Trade (PAT) scheme is a market based mechanism to optimize energy efficiency in industries. This mechanism allows plants to trade energy efficiency certificates in the market.

Karnataka ranks seventh in the production of cement in the country with an annual production of 12.1 million tons[30] of cement which emits 7.6 million tons[31] of CO2 annually. Karnataka is also the third largest steel producer in India with an annual production of 10.7 million tons[32]. In Karnataka these two industries account for over 20% of the overall emissions of the state and over 40% of the emissions due from industrial sector. Future consumption, of cement and iron & steel needed, to sustain economic growth of the economy can be projected using elasticity of these resources to GDP growth. These elasticities have been estimated to be a little over 1. The emissions in 2020-21 are likely to increase to 54 million tons from 18 million tons for the year 2008-09.

- Most industries consume thermal and electrical energy. It is the requirement of thermal energy that can be reduced by adopting energy efficient practices.
- In the case of cement sector, adding fly ash and slag to clinker would reduce the carbon footprint considerably.
- Use of natural gas based DRI steel making process where possible as this has approximately 40% less CO2 emissions compared to a blast furnace (BOF) process[33].
- The use of solar thermal technologies for process heat applications can reduce the fossil fuel usage in industrial applications.

[30] Directorate of Economics and Statistics, Government of Karnataka
[31] Cement Manufacturers Association and MoEF – Assumptions -0.537 t CO2/t Clinker produced and production clinker uses 85% of total energy used in cement manufacturer.
[32] Karnataka advantage, steel sector ; http://www.advantageKarnataka.com/pdf/steel.pdf
[33] *Options for the Swedish steel industry – Energy efficiency measures and fuel conversation*, Linkoping University Post press

Households

Currently biomass is the mainstay as a cooking fuel in a large percentage of rural households. This results in deforestation and GHG emissions. Over 75% of the households in India use biomass. While biomass is considered carbon neutral their CH_4 and N_2O emissions are taken into account. Extending the supply of LPG to all households can have positive environmental impact and will help alleviate air pollution related health problems to households.

Ease of Implementation*

Cost		Moderately Challenging	Challenging	Very Challenging
Cost	**Negative**	• Energy efficient appliances • Energy efficiency equipment • Solar water heaters in residential, public and private buildings • Mileage standard in automobiles	• Energy efficiency in iron & steel industry.	• Clinker substitute by fly-ash and slag in cement • LED lighting • Efficient cook stoves • Efficient irrigation pumps in agriculture
	Modest	• Small hydro • Wind • Solar thermal applications for industrial heat, agricultural drying, water heating. • Supercritical technology • Nuclear	• Reduction in T&D losses • Building efficiency	• Efficiency from other sectors
	High	• Solar photovoltaic for electricity generation • Public bus-based transportation systems • Shift to freight by rail	• CSP • Shift to gas based (direct reduced iron) DRI in steel manufacturing	• Large hydro • Large scale deployment of CSP • Ultra supercritical coal technology

* Ease of Implementation based on financing issues, regulating support, agency issues, entrenched behaviour, supply constraints & technological readiness.

Sources: McKinsey Report on Environment and Energy Sustainability and CSTEP's analysis

Cost of Mitigation

The investments required for abatement and cost per ton of saved CO2 should be ideally obtained by undertaking a life cycle analysis of each option. However, due to lack of data this has not been done here. The economics of abatement option has been done to a reasonable accuracy. However, in this phase of the report we have not been able to look at the economics of mitigation of other sectors and options in particular transportation and industry. It is because these entail more complex analysis and data availability curtailed from this analysis.

For some of the abatement options such as in the power sector, the approximate capital investment was known and this was used (cost of operations and maintenance was ignored) to compute the cost of abatement. This analysis is given in table 2. As can be seen here, the abatement choices are compared to subcritical coal technology. As can be seen, the changeover to natural gas for generation of electricity has a negative cost for abatement since the capital cost here is lower than that of sub-critical technology. Here for the sake of analysis, it is assumed that all the planned supercritical and gas based plants are built.

In the case of compact fluorescent light (CFL) life cycle analysis, the switch over to CFL from the incandescent lamp would result in a negative cost of over Rs 4000 (profit). In the state of Karnataka, assuming that on average each urban household is given 2 CFLs the total investments today would be Rs 176 crores. However, the total amount saved due to reduced electricity cost would be Rs 949 crores. However, a note of caution, the CFL lamps contain mercury and hence proper disposal should be designed. Light Emitting Diodes (LED) have a negative abatement cost as well. However, this is not yet commercially viable.

Table 5: Mitigation Options for Power Section

Option	**Total New Capacity between 2010-2020(MW)**	**Net Generation (Billion kWh)**	**Cost Rs(crores)/ MW**	**Total Investments (crores)**	**CO2 savings vs. sub-critical (kg/kWh)**	**CO2 Annual Savings (Million Tons)**
Sub critical			4.5			
Super critical*	4,000	25.79	4.7	18,800	0.10	2.58
Gas*	3,500	14.90	2.7	9,450	0.48	7.15
Nuclear	685	4.32	7.0	4,792	0.93	4.02
Wind	552	0.95	9.0	4,968	0.93	0.88
Solar	177	0.30	14	2,478	0.93	0.28

* Under the assumption that all the super-critical and gas based plants under plans are built by 2020.

Assumptions in computing net generation

1. Plant load factors (PLF) for 2020: Coal 80%, gas and diesel 55%. Nuclear 80%, hydro 35 %, wind 17%, solar 20%, biomass and others 40%, Non utility (40%)
2. Auxiliary power consumption: coal 8%, nuclear 10.5%, gas and diesel 3.1%, hydro 0.5%, wind 2%, solar photovoltaic 1 %, concentrated solar thermal power 7%, biomass and others 7%, non utility (3%)
3. Specific emission of the total current fleet of coal and lignite power plants is 1.1 kg of CO_2 per net kWh. Based on CEA data (Central Electricity Authority, Co_2 baseline database, version 5 and November 2009).
4. For the new 500 MW sub critical power plants , net heat rate is 2450 Kcal/Kwh leading to a specific emission of 0.93 kg of CO_2 per net kWh (Discussion with NTPC and Future of Coal MIT report).For the super critical plants we have assumed a net heat rate of 2235 kCal per kWh leading to a specific emission of 0.83 kg of CO_2 per net kWh (Discussions with NTPC and Future of Coal, MIT report)
5. Non Utility CO_2 emissions are mainly based on diesel. Specific CO_2 emissions of 0.67 kg/kWh Net.